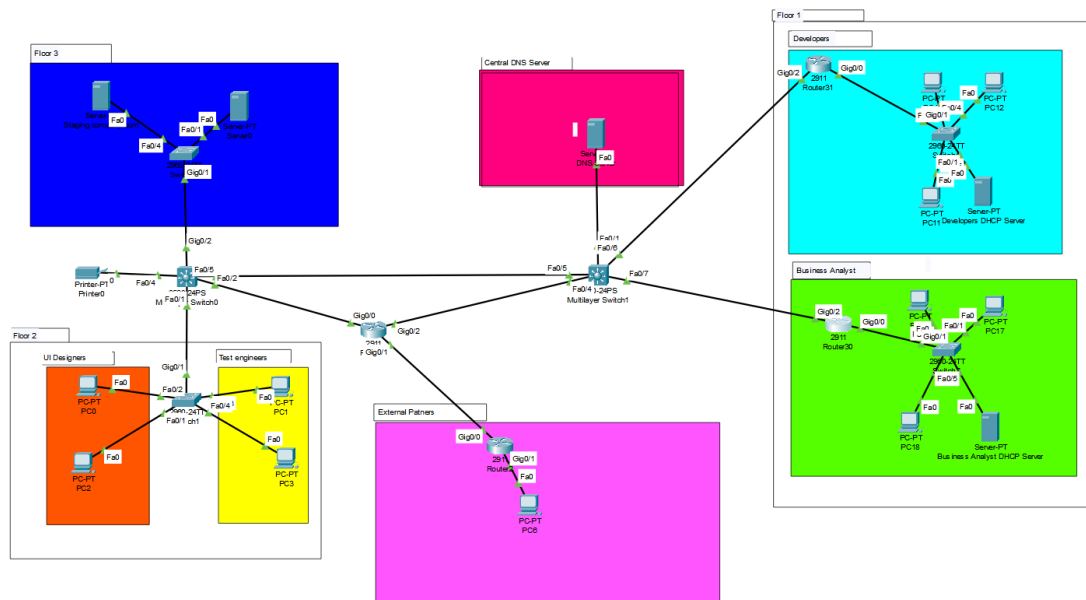


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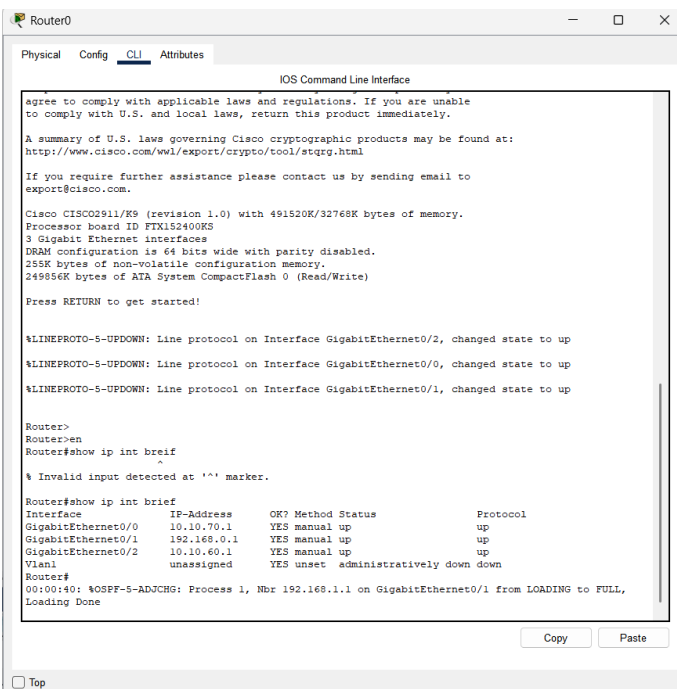
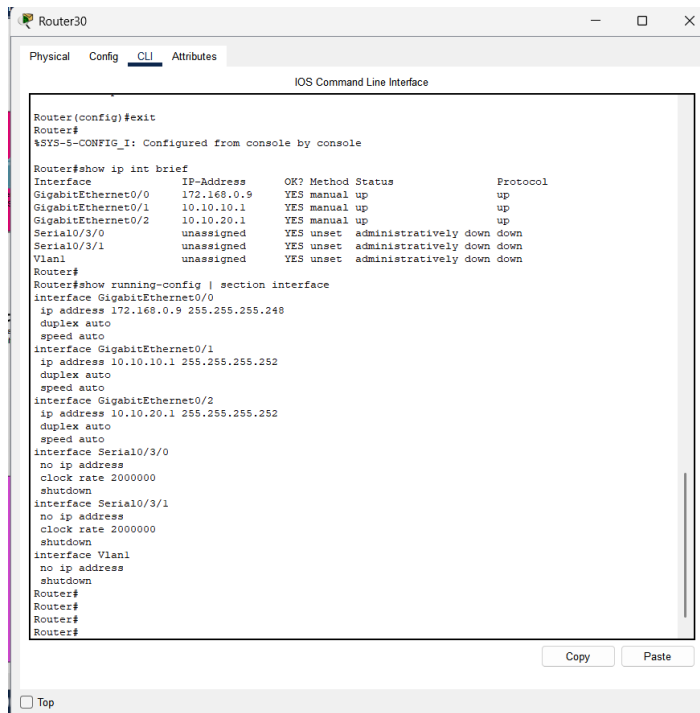
Question 1

1.1) Network Topology



Show ip int brief for the three routers

```
Router31
Router#
Router#
Router#
Router#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  172.168.0.1    YES manual up          up
GigabitEthernet0/1  10.10.30.1     YES manual up          up
GigabitEthernet0/2  10.10.40.1     YES manual up          up
Serial0/3/0        unassigned     YES unset  administratively down down
Serial0/3/1        unassigned     YES unset  administratively down down
Vlan1             unassigned     YES unset  administratively down down
Router#show running-config | section interface
interface GigabitEthernet0/0
ip address 172.168.0.1 255.255.255.248
duplex auto
speed auto
interface GigabitEthernet0/1
ip address 10.10.30.1 255.255.255.252
duplex auto
speed auto
interface GigabitEthernet0/2
ip address 10.10.40.1 255.255.255.252
duplex auto
speed auto
interface Serial0/3/0
no ip address
clock rate 2000000
shutdown
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
interface Vlan1
no ip address
shutdown
Router#
Router#
Router#
Router#
Router#
```



Show running config | section interface on all routers

The screenshot displays three Cisco Packet Tracer windows showing the running configuration for the 'interface' section of three routers: Router0, Router31, and Router30.

Router0 Configuration:

```
Router>
Router#
00:00:40: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
00:00:45: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/2 from LOADING to FULL, Loading Done
Router#show running-config | section interface
interface GigabitEthernet0/0
ip address 10.10.70.1 255.255.255.252
ip access-group ACL_OUTSIDE_IN in
ip nat inside
duplex auto
speed auto
interface GigabitEthernet0/1
ip address 192.168.0.1 255.255.255.248
ip nat outside
duplex auto
speed auto
interface GigabitEthernet0/2
ip address 10.10.40.1 255.255.255.252
ip nat inside
duplex auto
speed auto
interface Vlan1

```

Router31 Configuration:

```
Router(config)#interface GigabitEthernet0/2
Router(config-if)#exit
Router(config)#
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show running-config | section interface
interface GigabitEthernet0/0
ip address 172.168.0.1 255.255.255.248
duplex auto
speed auto
interface GigabitEthernet0/1
ip address 10.10.30.1 255.255.255.252
duplex auto
speed auto
interface GigabitEthernet0/2
ip address 10.10.40.1 255.255.255.252
duplex auto
speed auto
interface Serial0/3/0
no ip address
clock rate 2000000
shutdown
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
interface Vlan1
no ip address

```

Router30 Configuration:

```
Router>en
Router#show running-config | section interface
interface GigabitEthernet0/0
ip address 172.168.0.9 255.255.255.248
duplex auto
speed auto
interface GigabitEthernet0/1
ip address 10.10.10.1 255.255.255.252
duplex auto
speed auto
interface GigabitEthernet0/2
ip address 10.10.20.1 255.255.255.252
duplex auto
speed auto
interface Serial0/3/0
no ip address
clock rate 2000000
shutdown
interface Serial0/3/1
no ip address
clock rate 2000000
shutdown
interface Vlan1
no ip address

```

The background shows a network diagram with a Central DNS Server connected to Router31. Router31 is connected to Router30 via GigabitEthernet0/2 and GigabitEthernet0/1. Router30 is connected to a PC1 and PC2 via FastEthernet0/1 and FastEthernet0/2. The network is running OSPF.

Show vlan brief

Note that VLANs weren't created on the Business analyst router and Developers router just the multilayer switch under Floor 2 and 3

The image displays a Cisco Packet Tracer simulation environment. It includes three CLI windows showing the output of the 'show vlan brief' command on different switches, and a network diagram showing the topology.

Switch0 CLI Output:

```
Switch>show vlan brief
VLAN Name      Status Ports
-----
1 default      active Fa0/1, Fa0/2, Fa0/3, Fa0/6
                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                Fa0/23, Fa0/24, Gig0/2
10 staging-environment active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
```

Multilayer Switch0 CLI Output:

```
Switch>show vlan brief
VLAN Name      Status Ports
-----
1 default      active
10 Staging-Environment active
20 UI-Designers active
30 Test-Engineers active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
```

Switch1 CLI Output:

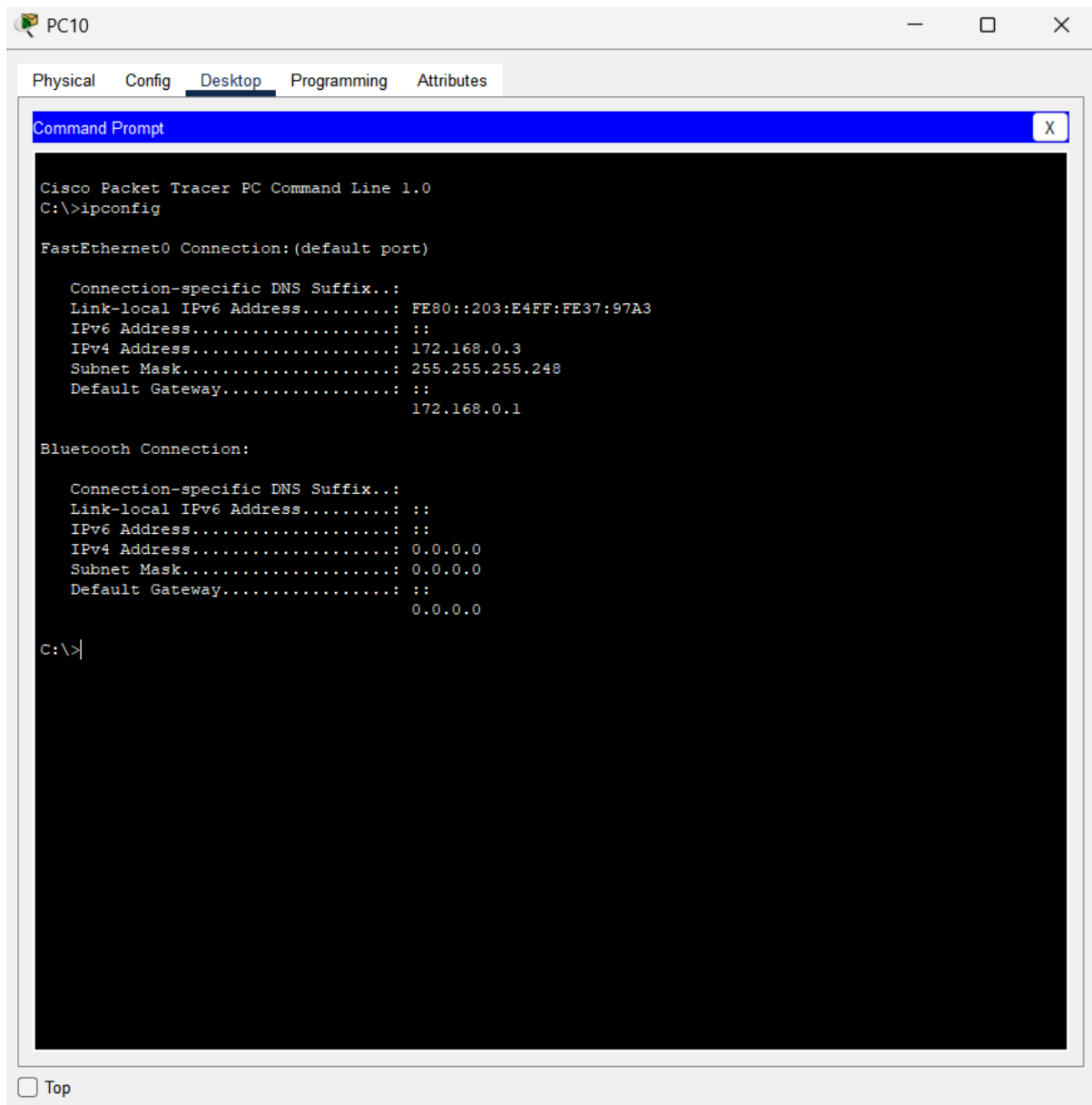
```
Switch>show vlan brief
VLAN Name      Status Ports
-----
1 default      active Fa0/5, Fa0/6, Fa0/7, Fa0/8
                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                Gig0/2
20 UI-Designers active Fa0/1, Fa0/2
30 Test-Engineers active Fa0/3, Fa0/4
90 printers active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
Switch#
```

Network Diagram:

The diagram shows a network topology with three main components: a Business Analyst Router (top left), a Developers Router (top right), and a Multilayer Switch (bottom center). The Business Analyst Router is connected to the Multilayer Switch via its GigabitEthernet0/1 interface. The Developers Router is connected to the Multilayer Switch via its GigabitEthernet0/1 interface. The Multilayer Switch is connected to two other switches, Switch0 and Switch1, via their GigabitEthernet0/1 interfaces. Switch0 is connected to a PC (PC1) and a server (Server1). Switch1 is connected to a PC (PC2) and a server (Server2). The Multilayer Switch is also connected to a PC (PC3) and a server (Server3) via its GigabitEthernet0/1 interface.

Running a ipconfig on the pc

Developers pc



Business analyst

The screenshot shows a Cisco Packet Tracer interface for PC19. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the output of the 'ipconfig' command, detailing network configuration for both FastEthernet0 and Bluetooth connections.

PC19

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::207:ECFF:FE66:8824
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.168.0.13
    Subnet Mask . . . . .: 255.255.255.248
    Default Gateway . . . . .: ::
                                172.168.0.9

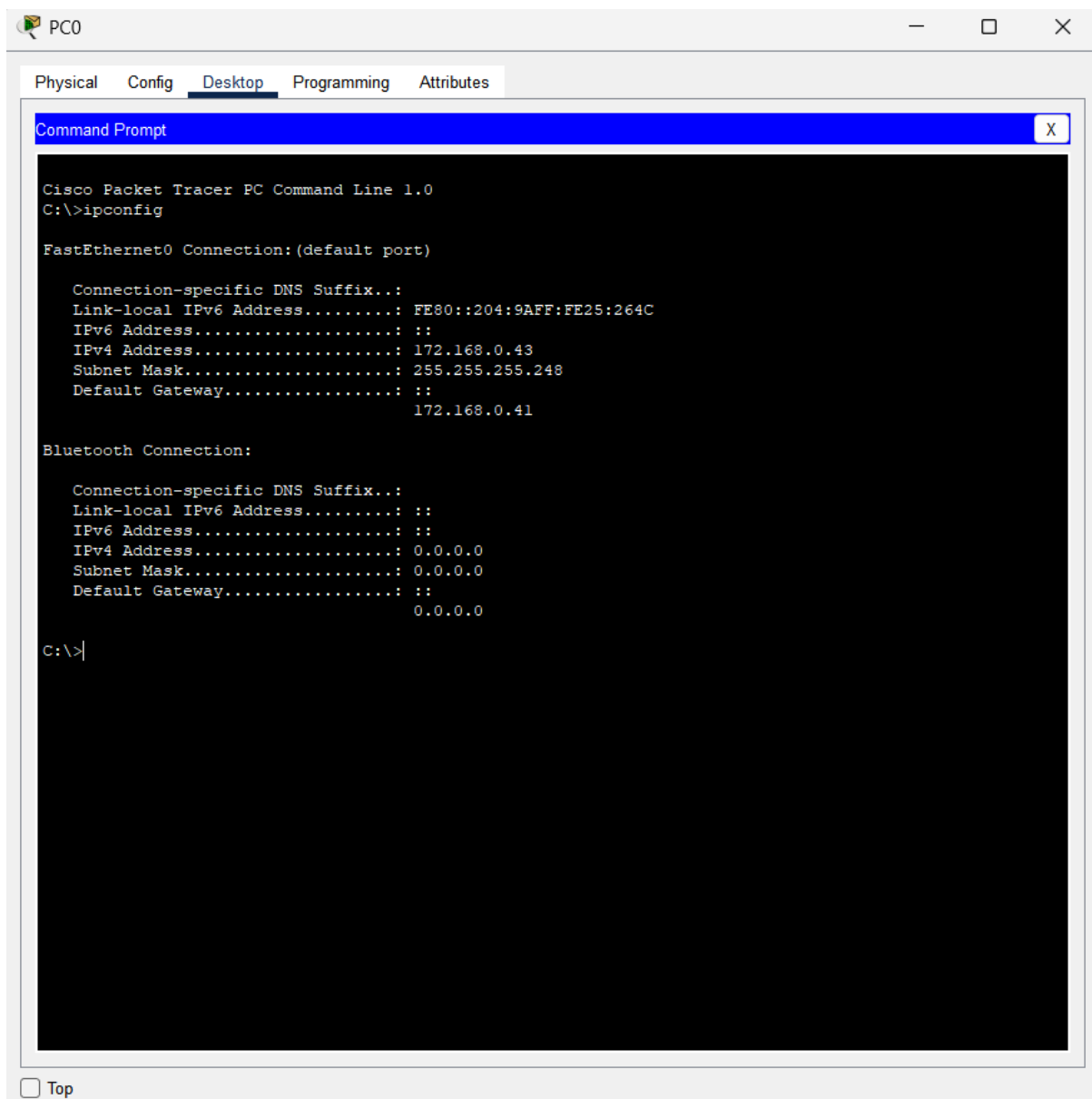
Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

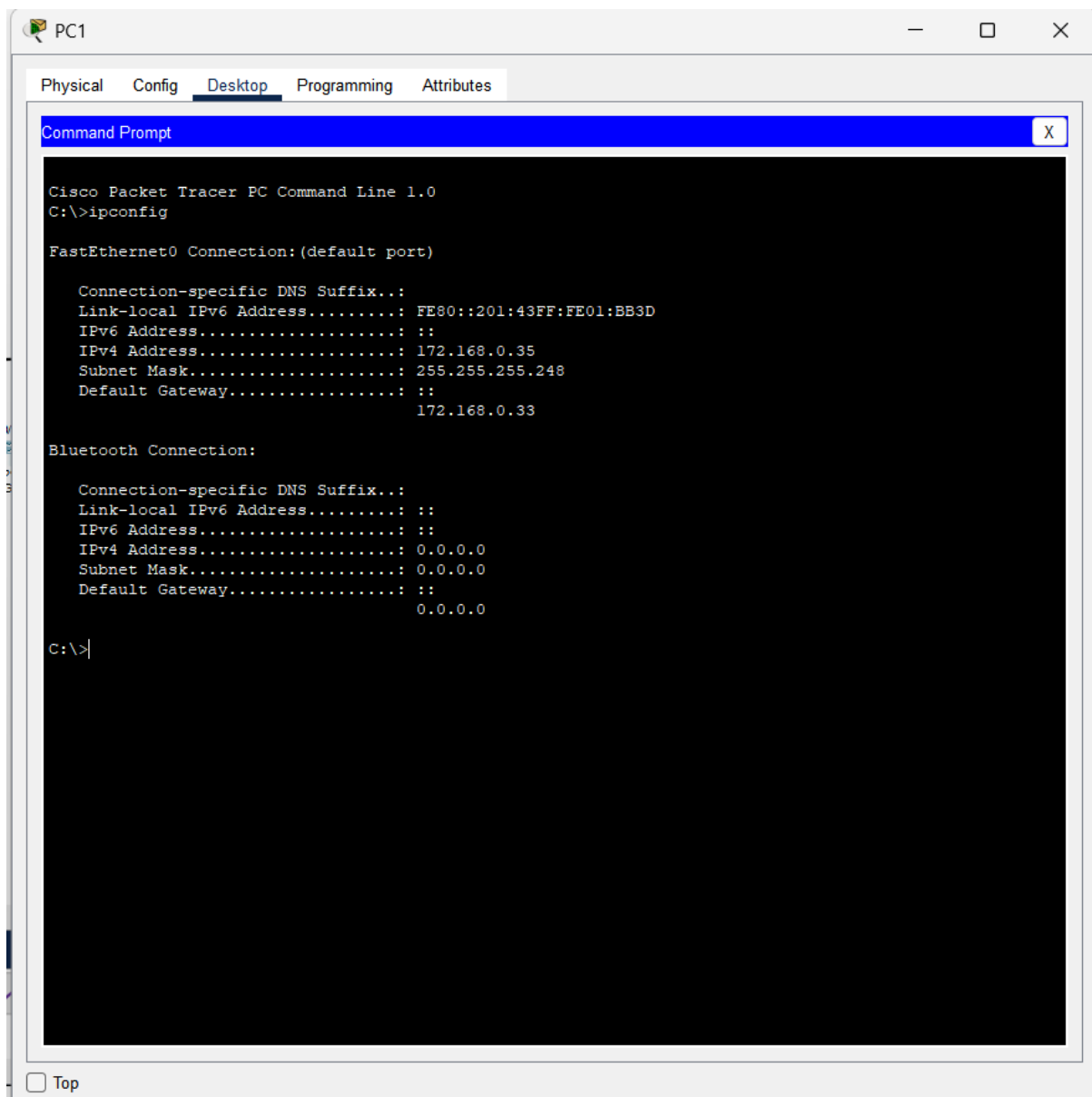
C:\>|
```

☐ Top

UI Designers



Test engineers



Staging server

Staging.company.com

PhysicalConfigServicesDesktopProgrammingAttributes

Command Prompt

X

Cisco Packet Tracer SERVER Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::202:4AFF:FE6A:53D8
IPv6 Address.....: ::
IPv4 Address.....: 172.168.0.18
Subnet Mask.....: 255.255.255.248
Default Gateway.....: ::
172.168.0.17

FastEthernet1 Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::290:CFF:FE6B:7E43
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0

C:\>

☐ Top

1.2)DHCP server configuration

Developers

Developers DHCP Server

Physical Config Services Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYNLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface FastEthernet0 Service On Off

Pool Name serverPool

Default Gateway 172.168.0.1

DNS Server 172.168.0.26

Start IP Address : 172 168 0 3

Subnet Mask: 255 255 255 248

Maximum Number of Users : 3

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	172.168.0.1	172.168.0.26	172.168.0.3	255.255.255.248	3	0.0.0.0	0.0.0.0

Top

Business analyst

Business Analyst DHCP Server

Physical Config Services Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYNLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface FastEthernet0 Service On Off

Pool Name serverPool

Default Gateway 172.168.0.9

DNS Server 172.168.0.26

Start IP Address : 172 168 0 11

Subnet Mask: 255 255 255 248

Maximum Number of Users : 3

TFTP Server: 0.0.0.0

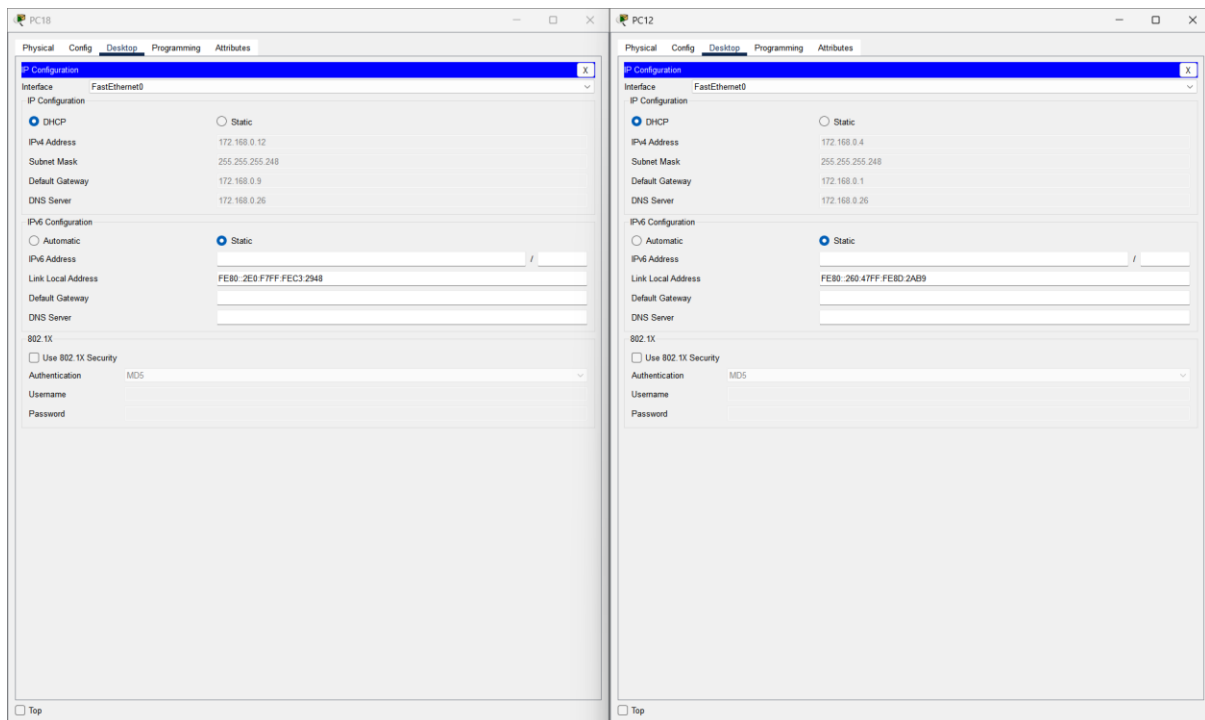
WLC Address: 0.0.0.0

Add Save Remove

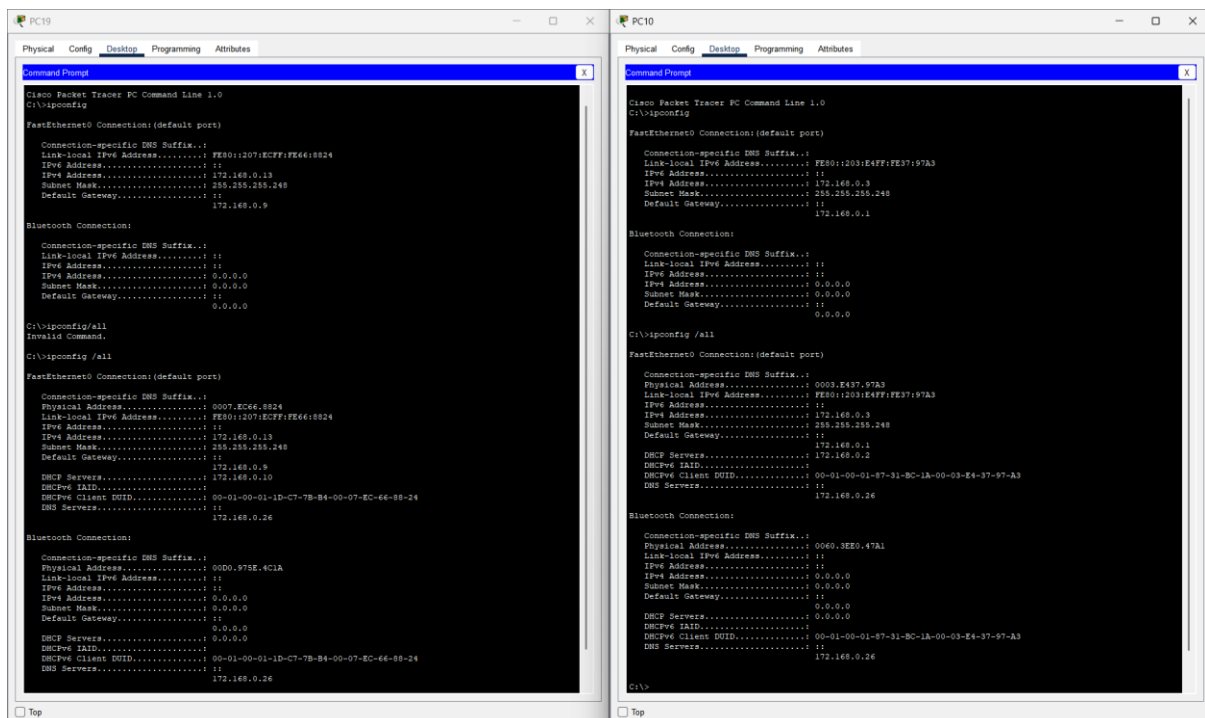
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	172.168.0.9	172.168.0.26	172.168.0.11	255.255.255.248	3	0.0.0.0	0.0.0.0

Top

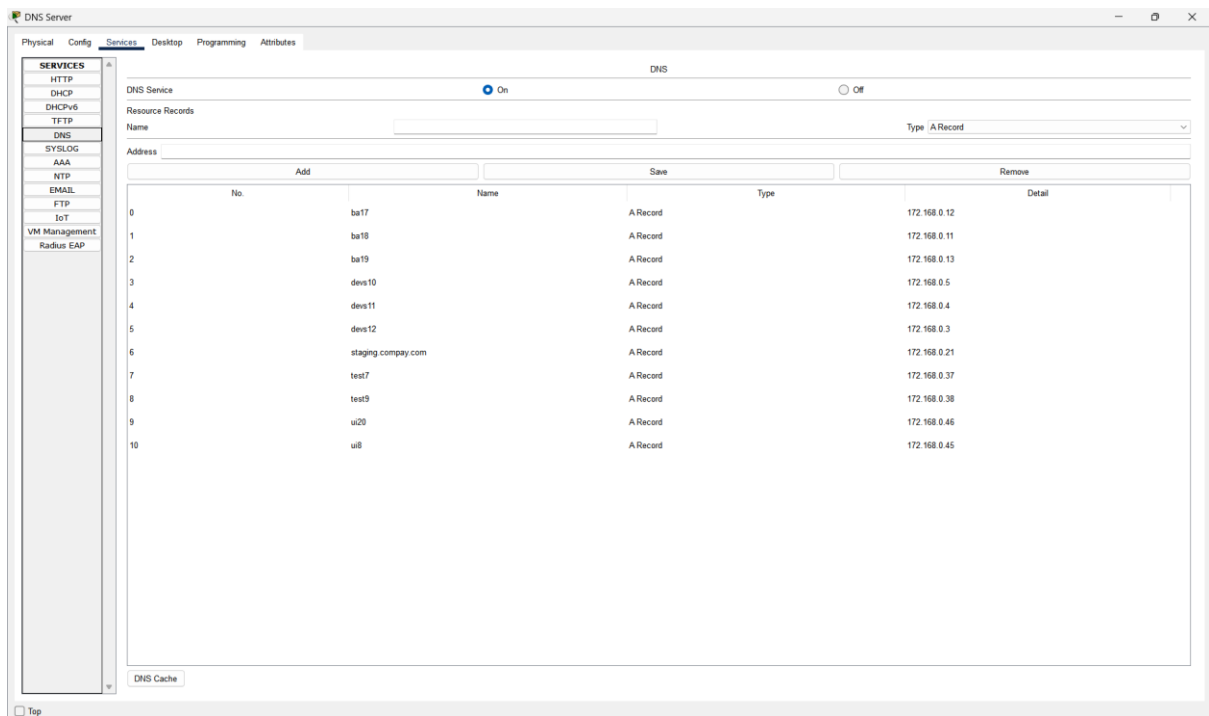
Verification of DHCP Working



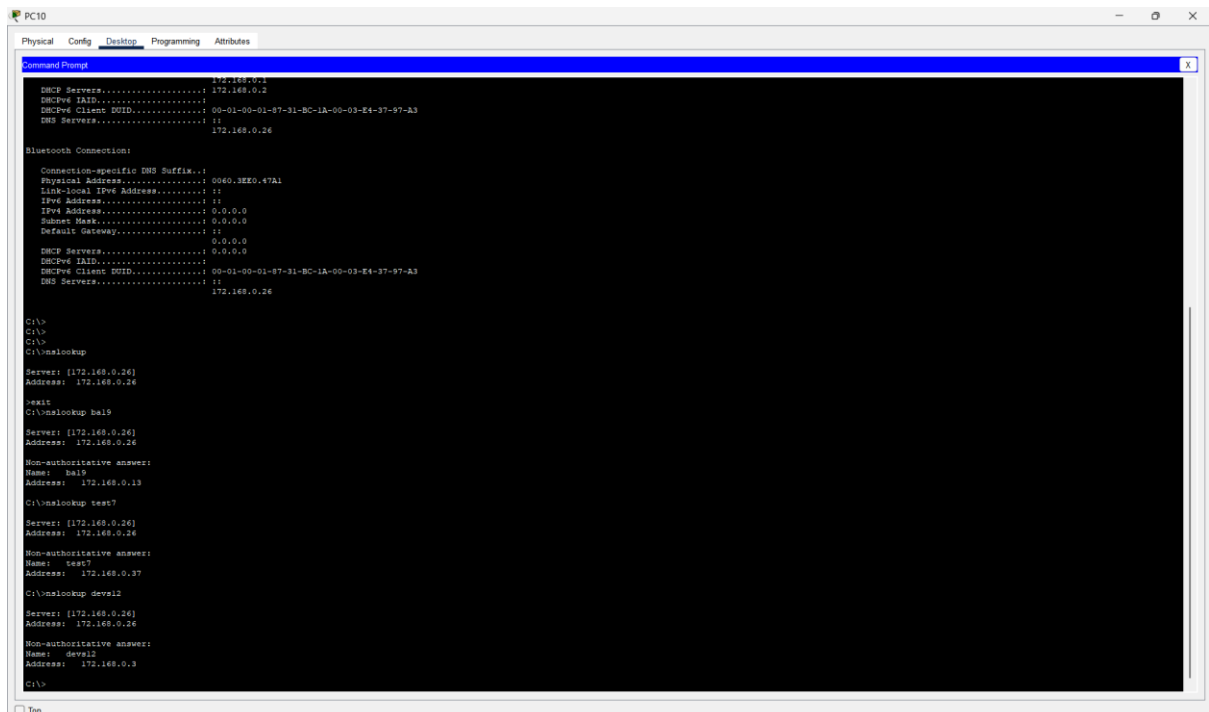
Ipnconfig all

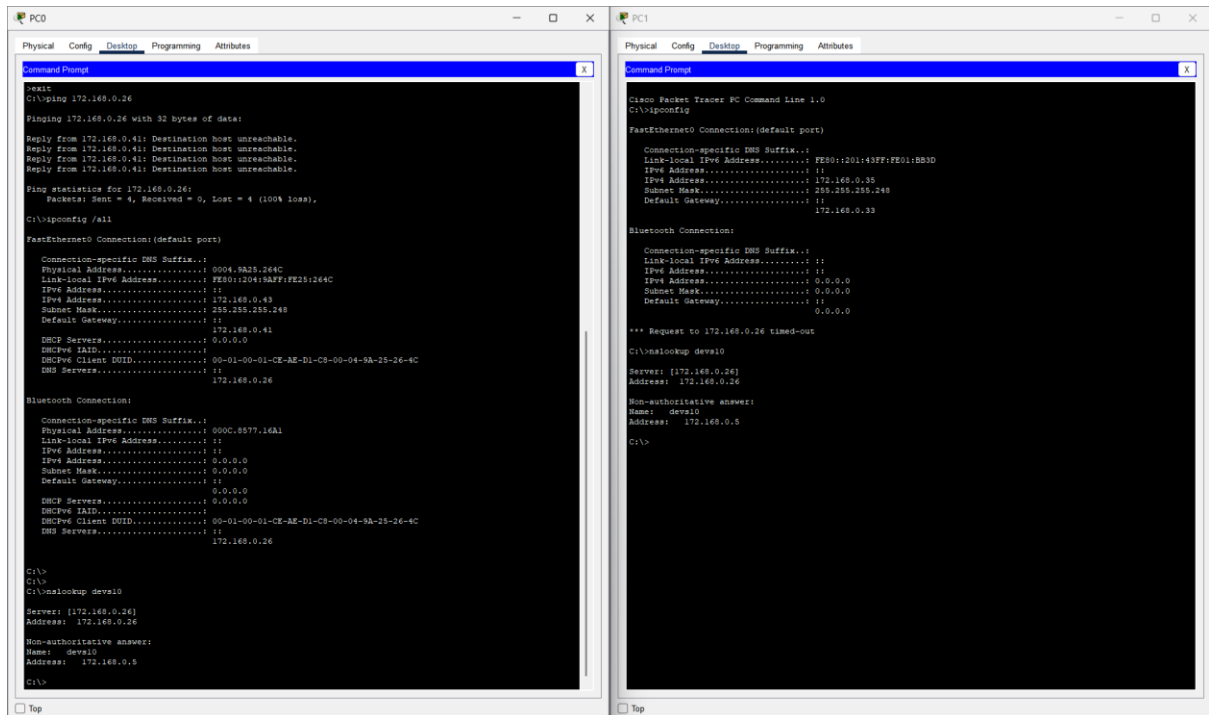
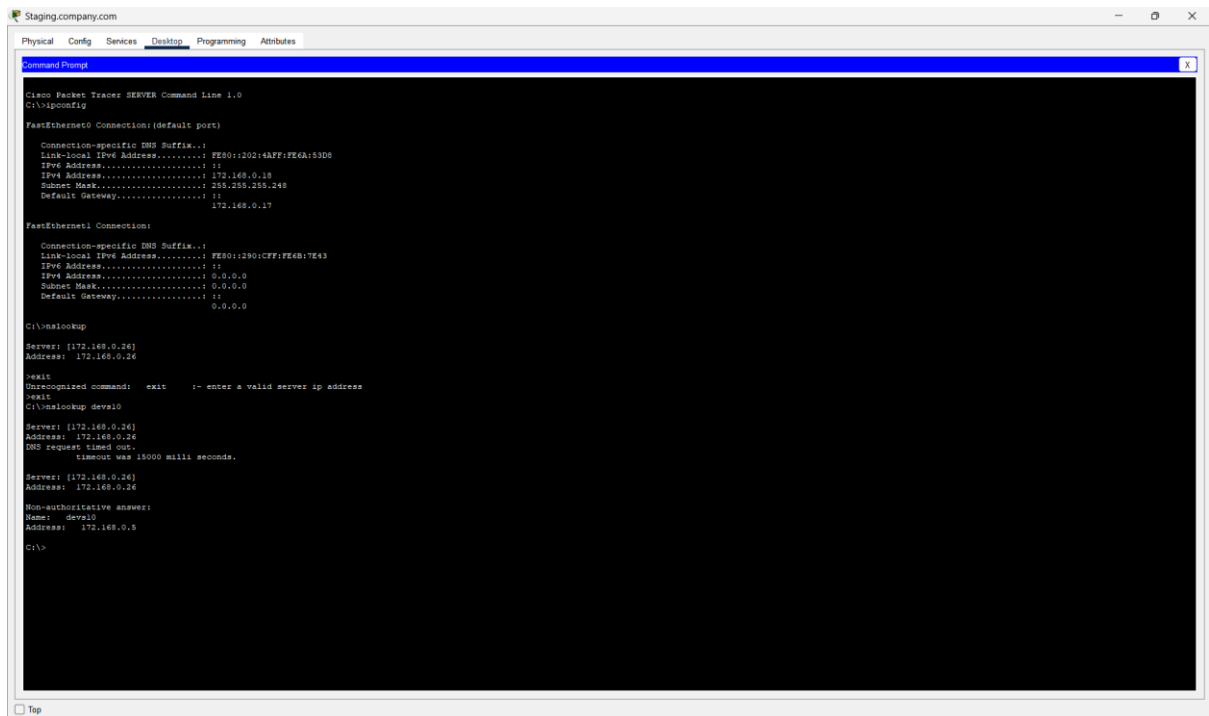


1.3)DNS server configuration

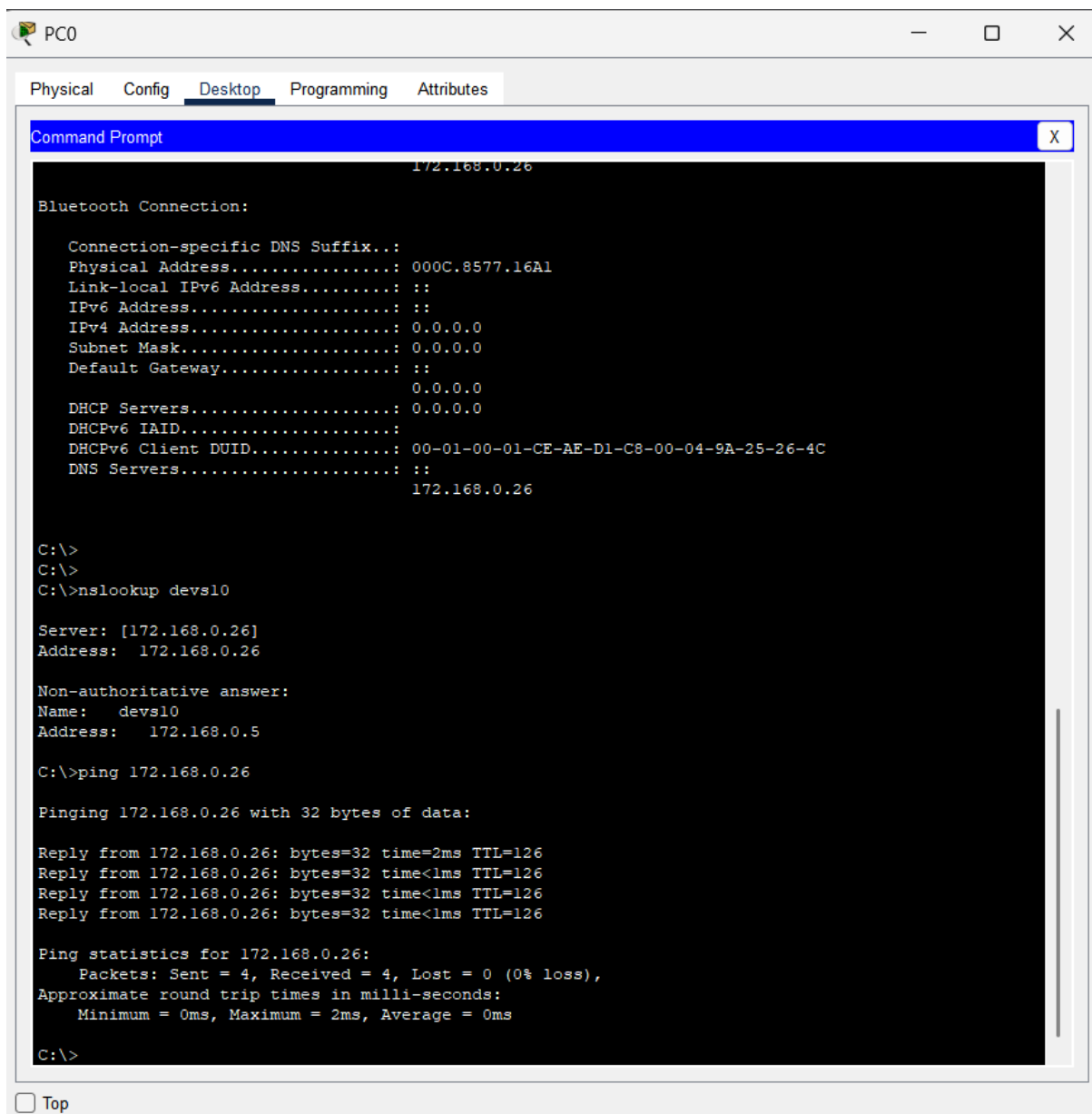


Nslookup from Pcs in the topology





Ping DNS server from UI designers



The screenshot shows a window titled "PC0" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the following output:

```
172.168.0.26

Bluetooth Connection:

Connection-specific DNS Suffix...:
Physical Address.....: 000C.8577.16A1
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
DHCP Servers.....: 0.0.0.0
DHCPv6 IAID.....:
DHCPv6 Client DUID.....: 00-01-00-01-CE-AE-D1-C8-00-04-9A-25-26-4C
DNS Servers.....: ::
172.168.0.26

C:\>
C:\>
C:\>nslookup devsl0

Server: [172.168.0.26]
Address: 172.168.0.26

Non-authoritative answer:
Name: devsl0
Address: 172.168.0.5

C:\>ping 172.168.0.26

Pinging 172.168.0.26 with 32 bytes of data:

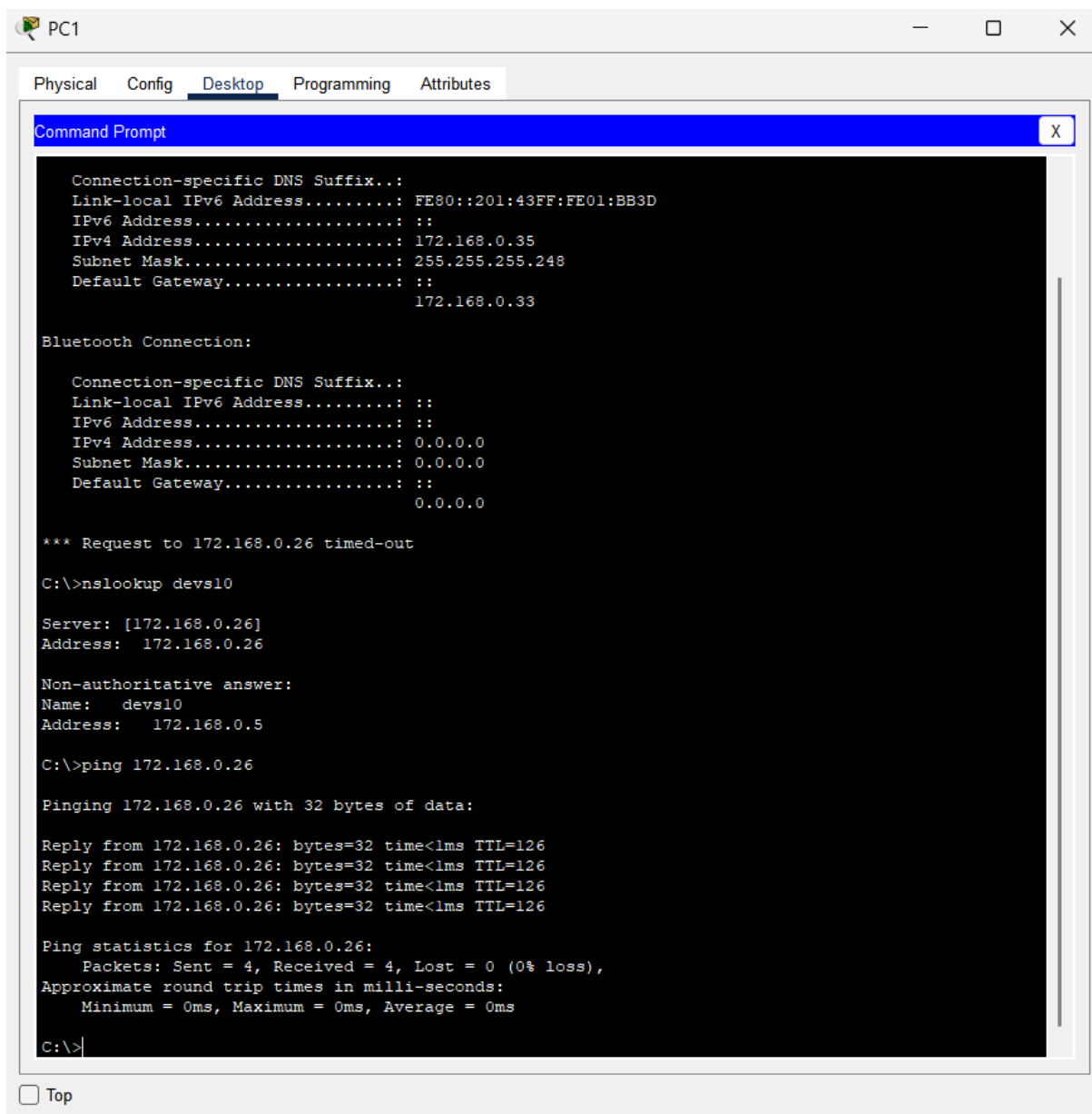
Reply from 172.168.0.26: bytes=32 time=2ms TTL=126
Reply from 172.168.0.26: bytes=32 time<1ms TTL=126
Reply from 172.168.0.26: bytes=32 time<1ms TTL=126
Reply from 172.168.0.26: bytes=32 time<1ms TTL=126

Ping statistics for 172.168.0.26:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

At the bottom of the window, there is a checkbox labeled "Top" which is currently unchecked.

Ping DNS server from test engineers



The screenshot shows a window titled "PC1" with tabs for Physical, Config, Desktop, Programming, and Attributes. The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the following output:

```
Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . .: FE80::201:43FF:FE01:BB3D
IPv6 Address . . . . .: ::
IPv4 Address . . . . .: 172.168.0.35
Subnet Mask . . . . .: 255.255.255.248
Default Gateway . . . . .: ::
                        172.168.0.33

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . .: ::
IPv6 Address . . . . .: ::
IPv4 Address . . . . .: 0.0.0.0
Subnet Mask . . . . .: 0.0.0.0
Default Gateway . . . . .: ::
                        0.0.0.0

*** Request to 172.168.0.26 timed-out

C:\>nslookup devsl0

Server: [172.168.0.26]
Address: 172.168.0.26

Non-authoritative answer:
Name:   devsl0
Address: 172.168.0.5

C:\>ping 172.168.0.26

Pinging 172.168.0.26 with 32 bytes of data:

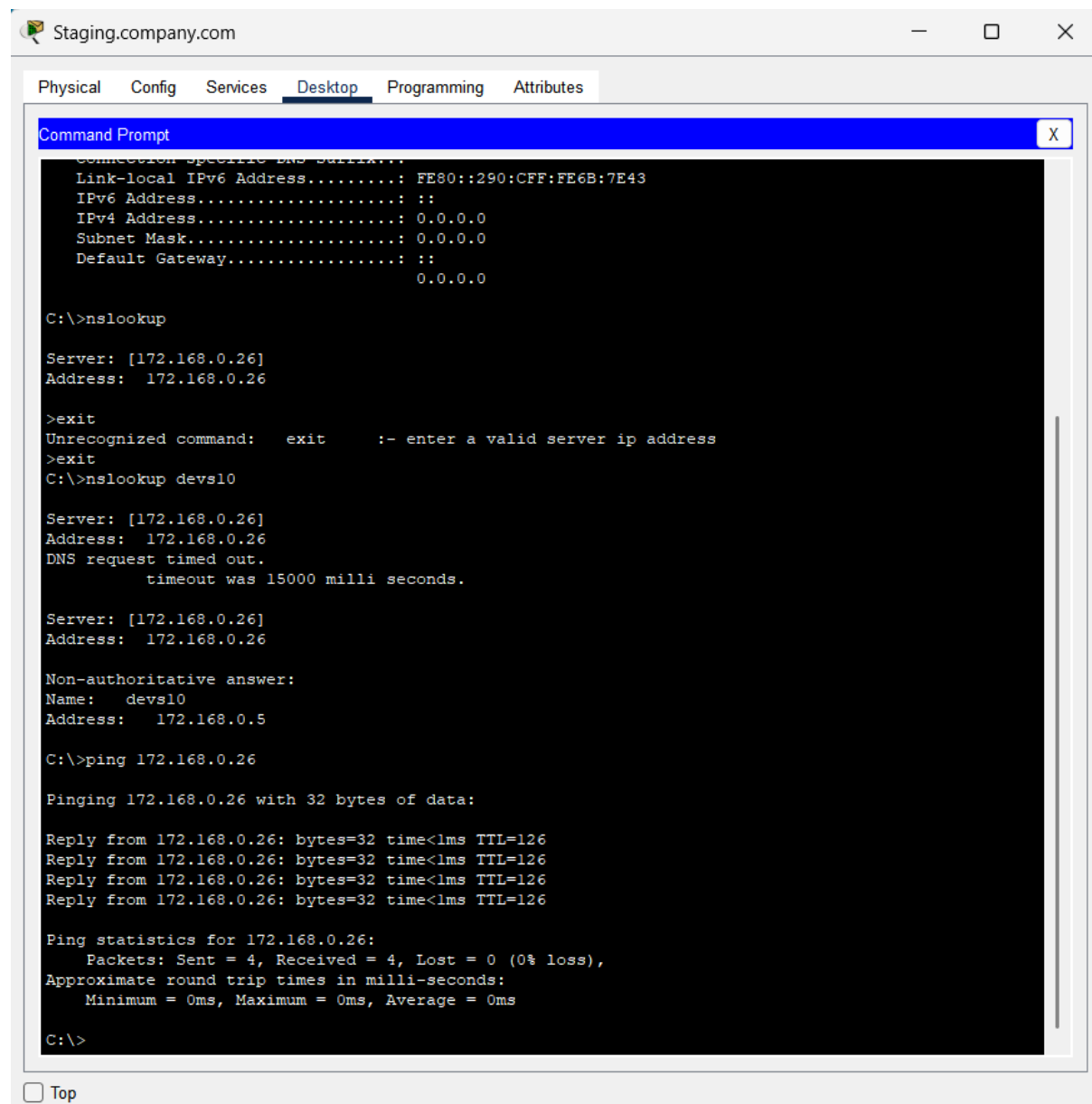
Reply from 172.168.0.26: bytes=32 time<lms TTL=126
Reply from 172.168.0.26: bytes=32 time<lms TTL=126
Reply from 172.168.0.26: bytes=32 time<lms TTL=126
Reply from 172.168.0.26: bytes=32 time<lms TTL=126

Ping statistics for 172.168.0.26:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

At the bottom left of the window, there is a checkbox labeled "Top".

Ping DNS server from staging environment



The screenshot shows a web browser window titled "Staging.company.com" with tabs for Physical, Config, Services, Desktop (selected), Programming, and Attributes. A Command Prompt window is open, displaying the following text:

```
Connection Specific DNS Settings...
Link-local IPv6 Address.....: FE80::290:CFF:FE6B:7E43
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
                        0.0.0.0

C:\>nslookup

Server: [172.168.0.26]
Address: 172.168.0.26

>exit
Unrecognized command:  exit      :- enter a valid server ip address
>exit
C:\>nslookup devsl0

Server: [172.168.0.26]
Address: 172.168.0.26
DNS request timed out.
        timeout was 15000 milli seconds.

Server: [172.168.0.26]
Address: 172.168.0.26

Non-authoritative answer:
Name:   devsl0
Address: 172.168.0.5

C:\>ping 172.168.0.26

Pinging 172.168.0.26 with 32 bytes of data:

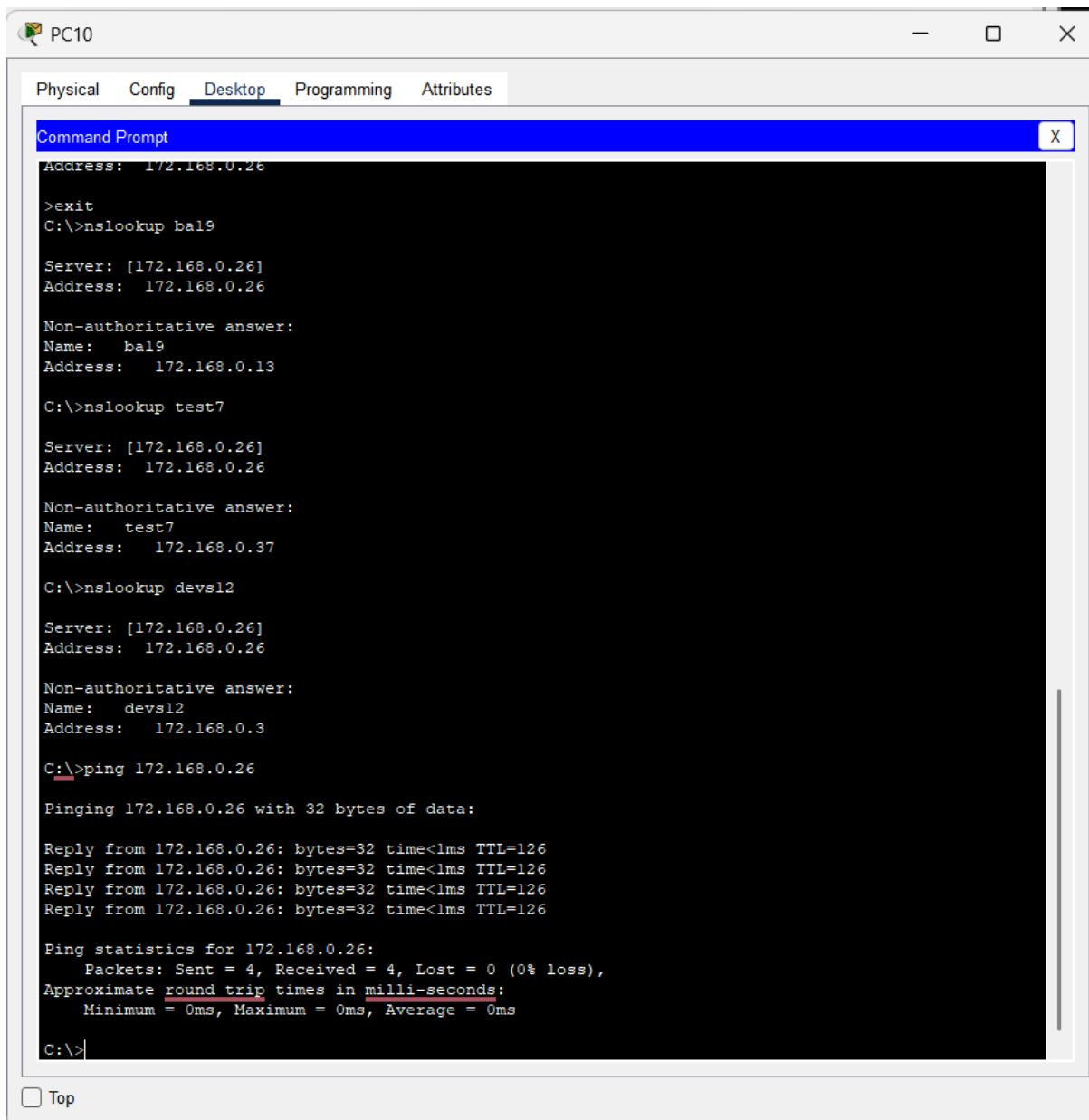
Reply from 172.168.0.26: bytes=32 time<lms TTL=126
Reply from 172.168.0.26: bytes=32 time<lms TTL=126
Reply from 172.168.0.26: bytes=32 time<lms TTL=126
Reply from 172.168.0.26: bytes=32 time<lms TTL=126

Ping statistics for 172.168.0.26:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

At the bottom left of the browser window, there is a checkbox labeled "Top".

Ping DNS server from floor 1



The screenshot shows a PC10 desktop environment with a window titled "PC10". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, and a "Command Prompt" window is open. The Command Prompt shows the following commands and output:

```
Address: 172.168.0.26
>exit
C:\>nslookup bal9

Server: [172.168.0.26]
Address: 172.168.0.26

Non-authoritative answer:
Name:    bal9
Address: 172.168.0.13

C:\>nslookup test7

Server: [172.168.0.26]
Address: 172.168.0.26

Non-authoritative answer:
Name:    test7
Address: 172.168.0.37

C:\>nslookup devsl2

Server: [172.168.0.26]
Address: 172.168.0.26

Non-authoritative answer:
Name:    devsl2
Address: 172.168.0.3

C:\>ping 172.168.0.26

Pinging 172.168.0.26 with 32 bytes of data:

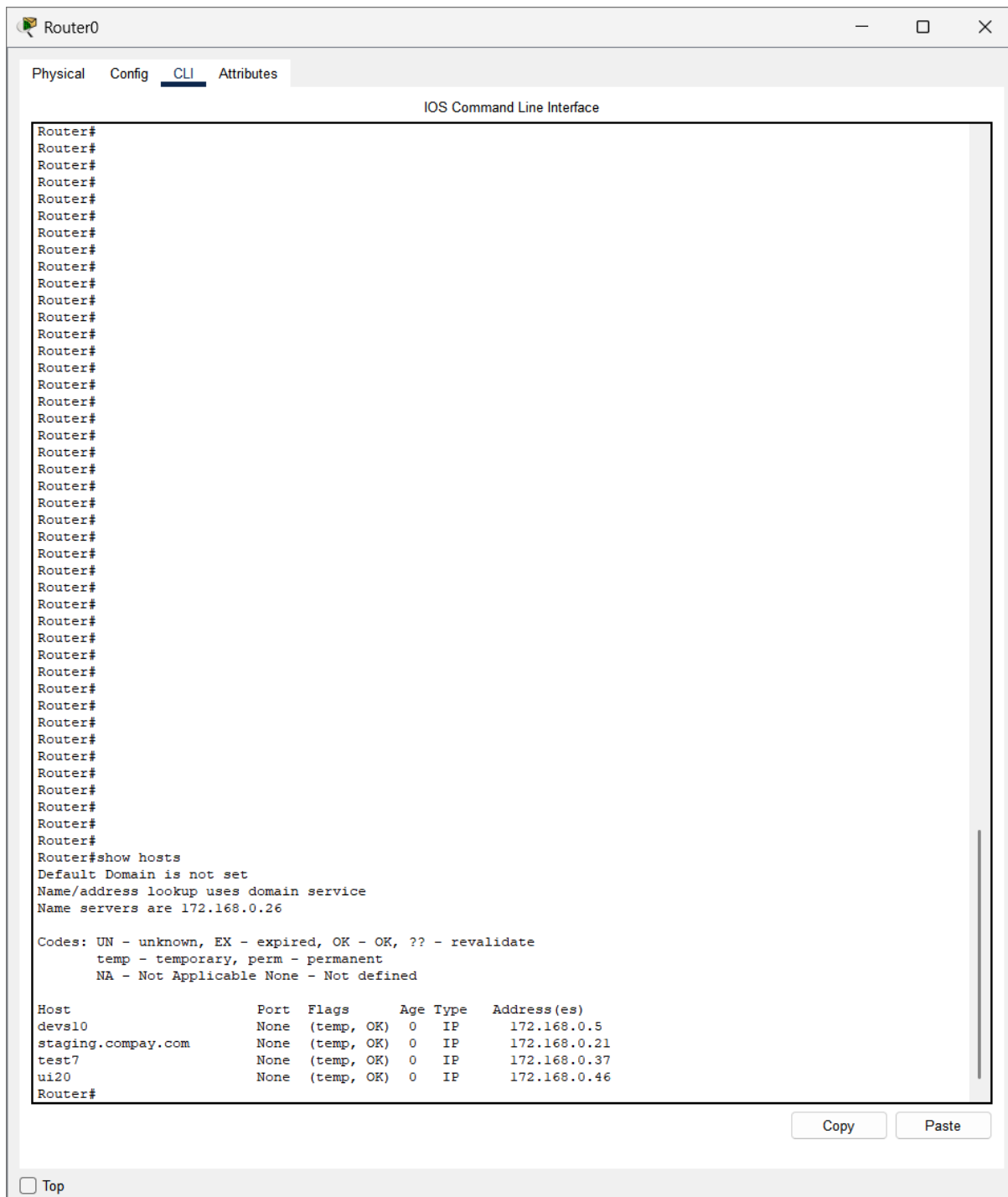
Reply from 172.168.0.26: bytes=32 time<lms TTL=126
Reply from 172.168.0.26: bytes=32 time<lms TTL=126
Reply from 172.168.0.26: bytes=32 time<lms TTL=126
Reply from 172.168.0.26: bytes=32 time<lms TTL=126

Ping statistics for 172.168.0.26:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

At the bottom of the window, there is a checkbox labeled "Top".

Show hosts



Question 2

2.1) Show vlan brief

Switch9

```
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gi0/0/2, Fa0/4, Fa0/5
10 staging-environment	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnnet-default	active	

Multilayer Switch0

```
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	
10 Staging-Environment	active	
20 UI-Designers	active	
30 Test-Engineers	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnnet-default	active	

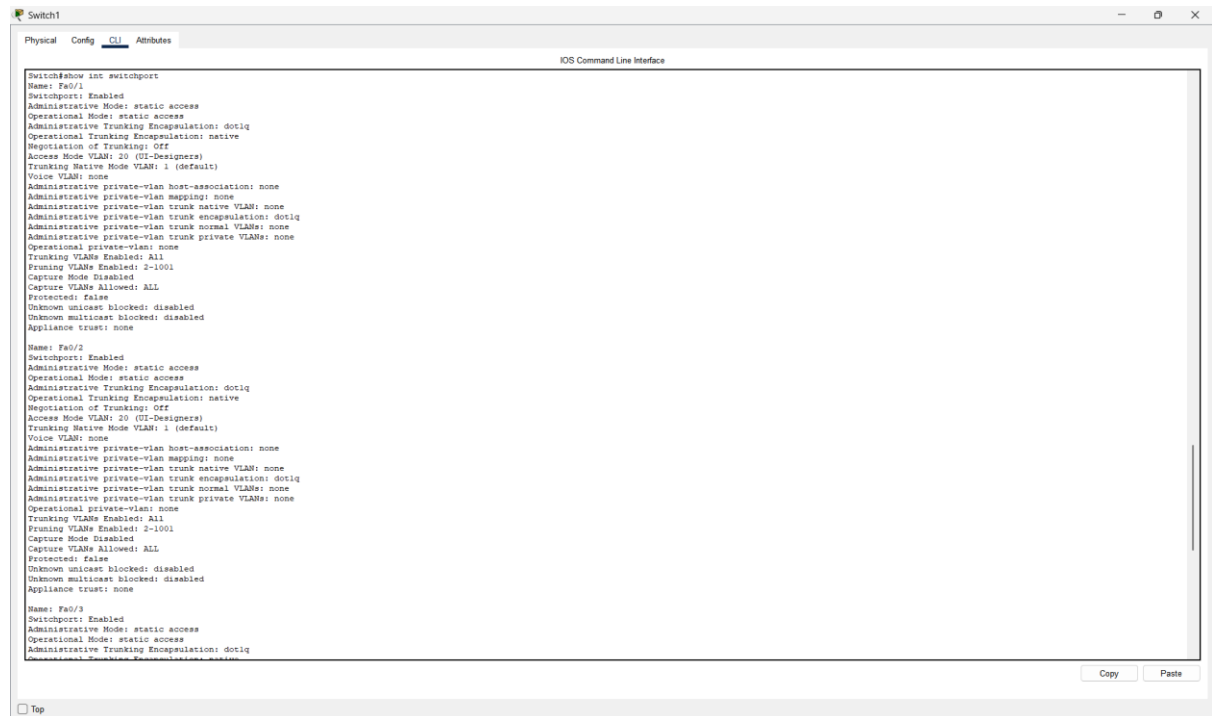
Switch1

```
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gi0/0/2
20 UI-Designers	active	Fa0/1, Fa0/2
30 Test-Engineers	active	Fa0/3, Fa0/4
80 printers	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnnet-default	active	

Show int switch port on Floor 2 switch

Part 1



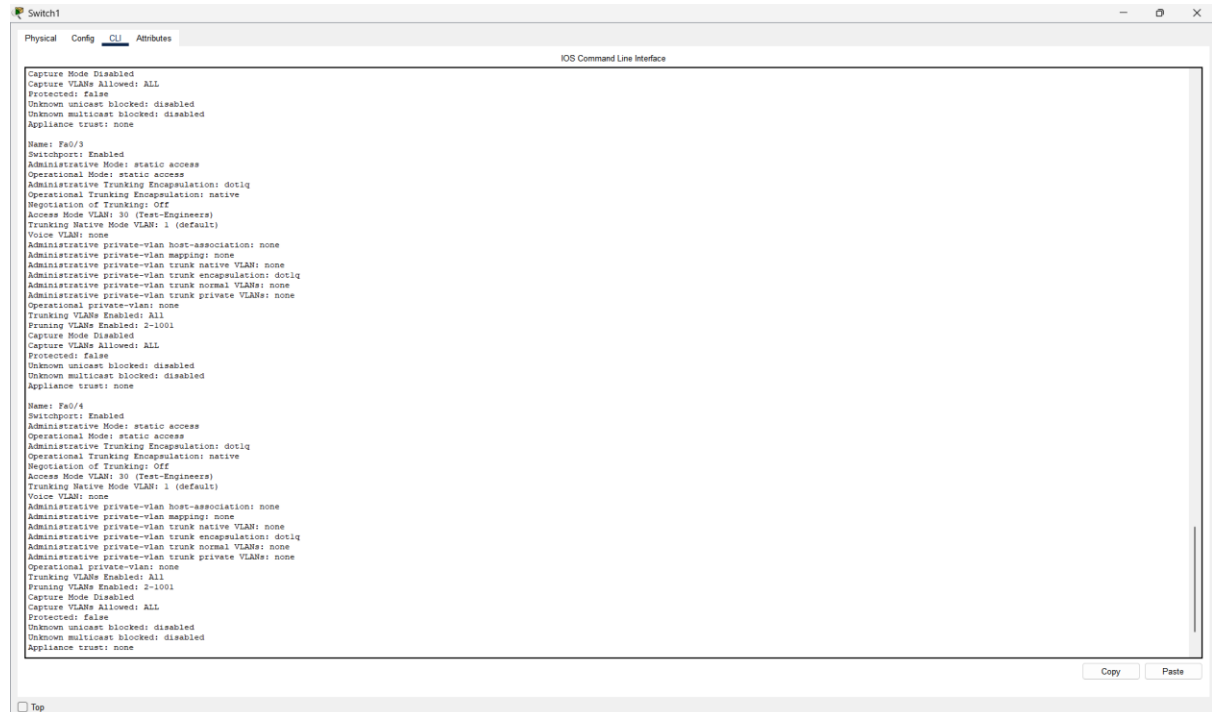
The screenshot shows a network switch CLI interface with the following configuration for ports Fa0/1, Fa0/2, and Fa0/3:

```
Switch1#show int switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 20 (UI-Designers)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none

Name: Fa0/2
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 20 (UI-Designers)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none

Name: Fa0/3
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 30 (Test-Engineers)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

Part 2



The screenshot shows a network switch CLI interface with the following configuration for ports Fa0/3, Fa0/4, and Fa0/5:

```
Switch1#show int switchport
Name: Fa0/3
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 30 (Test-Engineers)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none

Name: Fa0/4
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 30 (Test-Engineers)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none

Name: Fa0/5
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 30 (Test-Engineers)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

Show int trunk

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Name: Gig0/2
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: down
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none

Switch#
Switch#
Switch#
Switch#show int trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    20,30,50

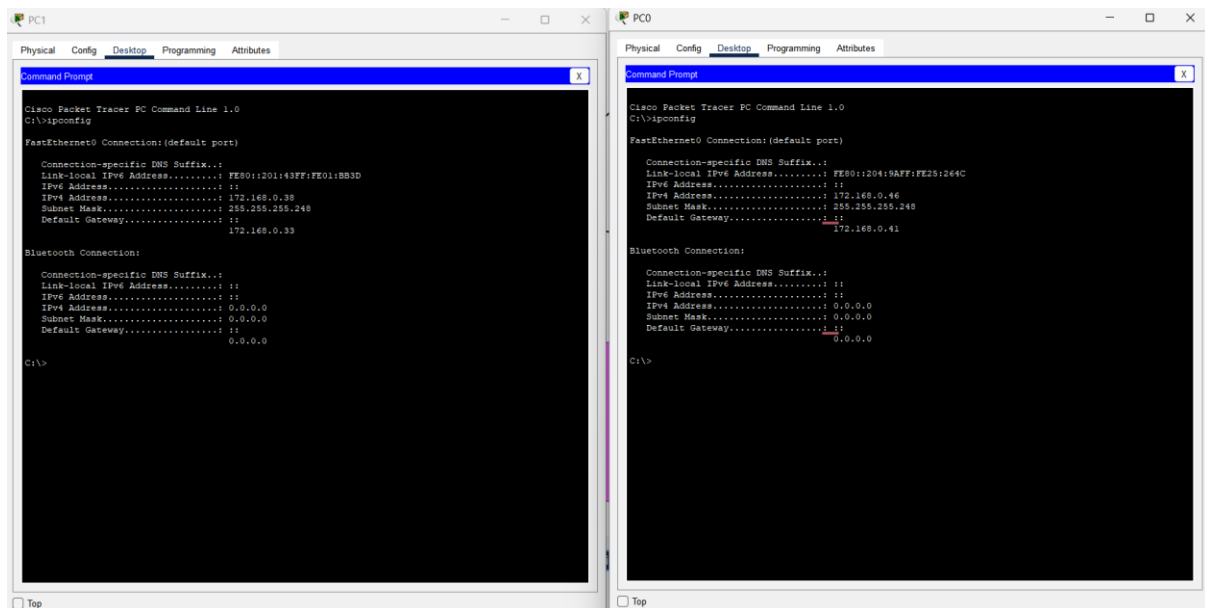
Port      Vlans allowed and active in management domain
Gig0/1    20,30,50

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    20,30,50
```

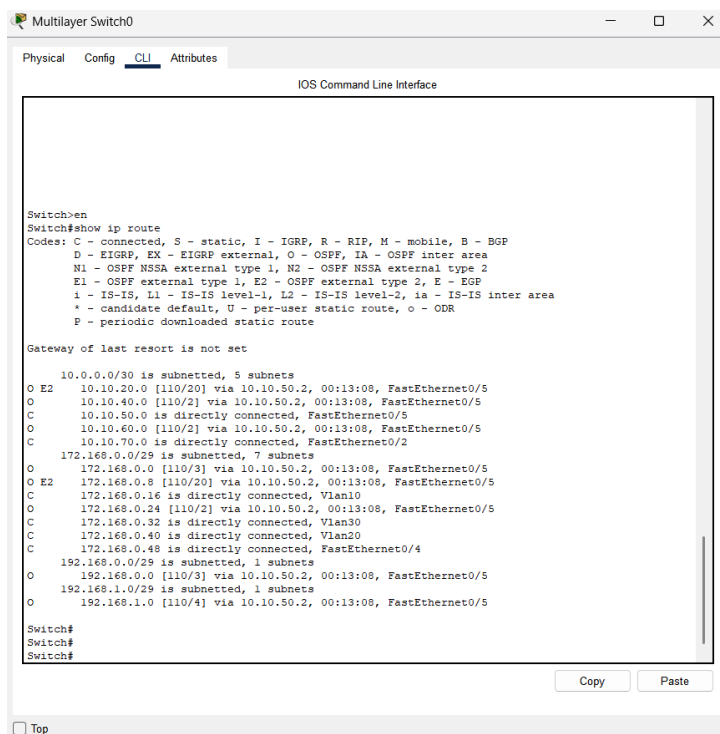
Copy Paste

☐ Top

Ipconfig on test engineers and UI designers PCs



2.2) Show ip route



Show ip int brief

Multilayer Switch0

Physical

Config

CLI

Attributes

IOS Command Line Interface

* invalid input detected at marker.

Switch#show ip int brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	10.10.70.2	YES	manual	up	up
FastEthernet0/3	unassigned	YES	unset	administratively down	down
FastEthernet0/4	172.168.0.51	YES	manual	up	up
FastEthernet0/5	10.10.50.1	YES	manual	up	up
FastEthernet0/6	unassigned	YES	unset	down	down
FastEthernet0/7	unassigned	YES	unset	down	down
FastEthernet0/8	unassigned	YES	unset	down	down
FastEthernet0/9	unassigned	YES	unset	down	down
FastEthernet0/10	unassigned	YES	unset	down	down
FastEthernet0/11	unassigned	YES	unset	down	down
FastEthernet0/12	unassigned	YES	unset	down	down
FastEthernet0/13	unassigned	YES	unset	down	down
FastEthernet0/14	unassigned	YES	unset	down	down
FastEthernet0/15	unassigned	YES	unset	down	down
FastEthernet0/16	unassigned	YES	unset	down	down
FastEthernet0/17	unassigned	YES	unset	down	down
FastEthernet0/18	unassigned	YES	unset	down	down
FastEthernet0/19	unassigned	YES	unset	down	down
FastEthernet0/20	unassigned	YES	unset	down	down
FastEthernet0/21	unassigned	YES	unset	down	down
FastEthernet0/22	unassigned	YES	unset	down	down
FastEthernet0/23	unassigned	YES	unset	down	down
FastEthernet0/24	unassigned	YES	unset	down	down
GigabitEthernet0/1	10.10.40.2	YES	manual	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	up	up
Vlan1	unassigned	YES	unset	up	up
Vlan10	172.168.0.17	YES	manual	up	up
Vlan20	172.168.0.41	YES	manual	up	up
Vlan30	172.168.0.33	YES	manual	up	up
Vlan40	unassigned	YES	unset	down	down
Vlan50	unassigned	YES	unset	down	down

Switch#

Switch#

Switch#show running-config | section interface Vlan

interface Vlan1

no ip address

interface Vlan10

mac-address 000c.85bb.b201

Copy

Paste

☐ Top

Show running-config | section interface Vlan

Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

FastEthernet0/18	unassigned	YES	unset	down	down
FastEthernet0/19	unassigned	YES	unset	down	down
FastEthernet0/20	unassigned	YES	unset	down	down
FastEthernet0/21	unassigned	YES	unset	down	down
FastEthernet0/22	unassigned	YES	unset	down	down
FastEthernet0/23	unassigned	YES	unset	down	down
FastEthernet0/24	unassigned	YES	unset	down	down
GigabitEthernet0/1	10.10.40.2	YES	manual	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	up	up
Vlan1	unassigned	YES	unset	up	up
Vlan10	172.168.0.17	YES	manual	up	up
Vlan20	172.168.0.41	YES	manual	up	up
Vlan30	172.168.0.33	YES	manual	up	up
Vlan40	unassigned	YES	unset	down	down
Vlan50	unassigned	YES	unset	down	down

```
Switch#
Switch#
Switch#show running-config | section interface Vlan
interface Vlan1
  no ip address
interface Vlan10
  mac-address 000c.85bb.b201
  ip address 172.168.0.17 255.255.255.248
  ip access-group ACL_TO_STAGING out
interface Vlan20
  mac-address 000c.85bb.b202
  ip address 172.168.0.41 255.255.255.248
  ip access-group VLAN_ISOLATION in
interface Vlan30
  mac-address 000c.85bb.b203
  ip address 172.168.0.33 255.255.255.248
  ip access-group VLAN_ISOLATION in
interface Vlan40
  mac-address 000c.85bb.b204
  no ip address
interface Vlan50
  mac-address 000c.85bb.b205
  no ip address
Switch#
Switch#
Switch#
Switch#
```

2.3) Show access list and show running-config | section access-list

```
Switch#show access-list
Extended IP access list ACL_TO_STAGING
 10 permit ip 172.168.0.0 0.0.0.7 172.168.0.16 0.0.0.7
 20 deny ip 172.168.0.8 0.0.0.7 172.168.0.16 0.0.0.7
 30 deny ip 172.168.0.32 0.0.0.7 172.168.0.16 0.0.0.7
 40 deny ip 172.168.0.40 0.0.0.7 172.168.0.16 0.0.0.7
 50 permit ip any 172.168.0.16 0.0.0.7
Extended IP access list ACL_FROM_STAGING
 10 deny ip 172.168.0.16 0.0.0.7 172.168.0.0 0.0.0.255
 20 permit ip any any
Extended IP access list BLOCK_PRINTER_TO_STAGING
 10 deny ip 172.168.0.48 0.0.0.7 172.168.0.16 0.0.0.7
 20 permit ip any any
Extended IP access list ACL_FROM_EXTERNAL_PARTNER
 10 deny ip 192.168.0.0 0.0.0.3 any
 20 permit ip any any
Extended IP access list ACL_PRINTER_TO_EXTERNAL
 10 deny ip 172.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
 20 deny ip 192.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
Extended IP access list VLAN_PRINTER_TO_FLOORS
 10 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52
 20 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52
 30 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7
 40 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7
Extended IP access list VLAN_ISOLATION
 10 deny ip 192.168.0.0 0.0.0.7 172.168.0.32 0.0.0.7
 20 deny ip 192.168.0.0 0.0.0.7 172.168.0.40 0.0.0.7
 30 deny ip 172.168.0.32 0.0.0.7 192.168.0.0 0.0.0.7
 40 deny ip 172.168.0.40 0.0.0.7 192.168.0.0 0.0.0.7
 50 deny ip 172.168.0.32 0.0.0.7 172.168.0.40 0.0.0.7
 60 deny ip 172.168.0.40 0.0.0.7 172.168.0.32 0.0.0.7
 70 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52
 80 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52
 90 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7
 100 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7
 110 permit ip 172.168.0.32 0.0.0.7 172.168.0.24 0.0.0.7
 120 permit ip 172.168.0.40 0.0.0.7 172.168.0.24 0.0.0.7

Switch#
Switch#show running-config | section access-list
ip access-list extended ACL_TO_STAGING
 permit ip 172.168.0.0 0.0.0.7 172.168.0.16 0.0.0.7
 deny ip 172.168.0.8 0.0.0.7 172.168.0.16 0.0.0.7
 deny ip 172.168.0.32 0.0.0.7 172.168.0.16 0.0.0.7
 deny ip 172.168.0.40 0.0.0.7 172.168.0.16 0.0.0.7
 permit ip any 172.168.0.16 0.0.0.7
ip access-list extended ACL_FROM_STAGING
 deny ip 172.168.0.16 0.0.0.7 172.168.0.0 0.0.0.255
 permit ip any any
ip access-list extended BLOCK_PRINTER_TO_STAGING
 deny ip 172.168.0.48 0.0.0.7 172.168.0.16 0.0.0.7
 permit ip any any
ip access-list extended ACL_FROM_EXTERNAL_PARTNER
 deny ip 192.168.0.0 0.0.0.3 any
 permit ip any any
ip access-list extended ACL_PRINTER_TO_EXTERNAL
 deny ip 172.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
 deny ip 192.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7

Switch#
Switch#
```

```
Switch#show running-config | section access-list
ip access-list extended ACL_TO_STAGING
 permit ip 172.168.0.0 0.0.0.7 172.168.0.16 0.0.0.7
 deny ip 172.168.0.8 0.0.0.7 172.168.0.16 0.0.0.7
 deny ip 172.168.0.32 0.0.0.7 172.168.0.16 0.0.0.7
 deny ip 172.168.0.40 0.0.0.7 172.168.0.16 0.0.0.7
 permit ip any 172.168.0.16 0.0.0.7
ip access-list extended ACL_FROM_STAGING
 deny ip 172.168.0.16 0.0.0.7 172.168.0.0 0.0.0.255
 permit ip any any
ip access-list extended BLOCK_PRINTER_TO_STAGING
 deny ip 172.168.0.48 0.0.0.7 172.168.0.16 0.0.0.7
 permit ip any any
ip access-list extended ACL_FROM_EXTERNAL_PARTNER
 deny ip 192.168.0.0 0.0.0.3 any
 permit ip any any
ip access-list extended ACL_PRINTER_TO_EXTERNAL
 deny ip 172.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
 deny ip 192.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
ip access-list extended VLAN_PRINTER_TO_FLOORS
 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52
 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7
 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7
ip access-list extended VLAN_ISOLATION
 deny ip 192.168.0.0 0.0.0.7 172.168.0.32 0.0.0.7
 deny ip 192.168.0.0 0.0.0.7 172.168.0.40 0.0.0.7
 deny ip 172.168.0.32 0.0.0.7 192.168.0.0 0.0.0.7
 deny ip 172.168.0.40 0.0.0.7 192.168.0.0 0.0.0.7
 deny ip 172.168.0.32 0.0.0.7 172.168.0.40 0.0.0.7
 deny ip 172.168.0.40 0.0.0.7 172.168.0.32 0.0.0.7
 deny ip 172.168.0.32 0.0.0.7 host 172.168.0.52
 deny ip 172.168.0.40 0.0.0.7 host 172.168.0.52
 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7
 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7
 permit ip 172.168.0.32 0.0.0.7 172.168.0.24 0.0.0.7
 permit ip 172.168.0.40 0.0.0.7 172.168.0.24 0.0.0.7
Switch#
Switch#
```

Hit count

Multilayer Switch0

Physical

Config

CLI

Attributes

IOS Command Line Interface

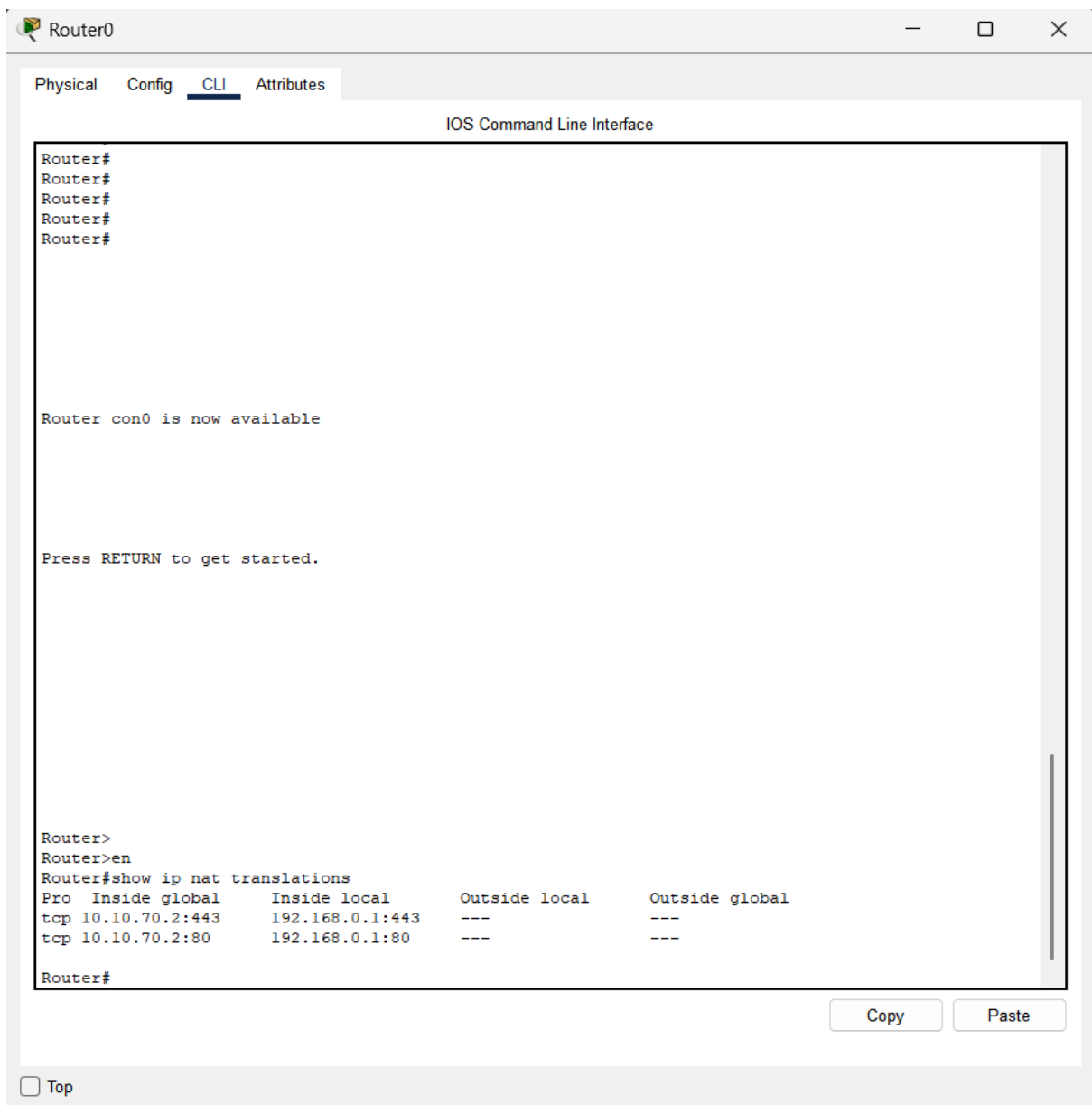
```
permit ip 172.168.0.32 0.0.0.7 172.168.0.24 0.0.0.7
permit ip 172.168.0.40 0.0.0.7 172.168.0.24 0.0.0.7
Switch#
Switch#
Switch#
Switch#show access-list
Extended IP access list ACL_TO_STAGING
 10 permit ip 172.168.0.0 0.0.0.7 172.168.0.16 0.0.0.7
 20 deny ip 172.168.0.8 0.0.0.7 172.168.0.16 0.0.0.7
 30 deny ip 172.168.0.32 0.0.0.7 172.168.0.16 0.0.0.7
 40 deny ip 172.168.0.40 0.0.0.7 172.168.0.16 0.0.0.7
 50 permit ip any 172.168.0.16 0.0.0.7
Extended IP access list ACL_FROM_STAGING
 10 deny ip 172.168.0.16 0.0.0.7 172.168.0.0 0.0.0.255
 20 permit ip any any
Extended IP access list BLOCK_PRINTER_TO_STAGING
 10 deny ip 172.168.0.48 0.0.0.7 172.168.0.16 0.0.0.7
 20 permit ip any any (3 match(es))
Extended IP access list ACL_FROM_EXTERNAL_PARTNER
 10 deny ip 192.168.0.0 0.0.0.3 any
 20 permit ip any any
Extended IP access list ACL_PRINTER_TO_EXTERNAL
 10 deny ip 172.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
 20 deny ip 192.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
Extended IP access list VLAN_PRINTER_TO_FLOOR3
 10 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52
 20 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52
 30 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7 (1 match(es))
 40 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7 (1 match(es))
Extended IP access list VLAN_ISOLATION
 10 deny ip 192.168.0.0 0.0.0.7 172.168.0.32 0.0.0.7
 20 deny ip 192.168.0.0 0.0.0.7 172.168.0.40 0.0.0.7
 30 deny ip 172.168.0.32 0.0.0.7 192.168.0.0 0.0.0.7
 40 deny ip 172.168.0.40 0.0.0.7 192.168.0.0 0.0.0.7
 50 deny ip 172.168.0.32 0.0.0.7 172.168.0.40 0.0.0.7 (1 match(es))
 60 deny ip 172.168.0.40 0.0.0.7 172.168.0.32 0.0.0.7 (1 match(es))
 70 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52 (1 match(es))
 80 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52 (2 match(es))
 90 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7
100 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7
110 permit ip 172.168.0.32 0.0.0.7 172.168.0.24 0.0.0.7
120 permit ip 172.168.0.40 0.0.0.7 172.168.0.24 0.0.0.7
```

Copy

Paste

☐ Top

3.1) Show ip nat translation on router



The screenshot shows a web-based interface for a router named Router0. The 'CLI' tab is selected, displaying the IOS Command Line Interface. The terminal output shows the user entering the command 'show ip nat translations' and receiving a table of active translations. The table has four columns: 'Pro', 'Inside global', 'Inside local', and 'Outside local'. The output shows two TCP translations from 10.10.70.2 to 192.168.0.1 on ports 443 and 80. There are also 'Copy' and 'Paste' buttons at the bottom right of the terminal area.

```
Router#  
Router#  
Router#  
Router#  
Router#  
  
Router con0 is now available  
  
Press RETURN to get started.  
  
Router>  
Router>en  
Router#show ip nat translations  
Pro  Inside global      Inside local    Outside local   Outside global  
tcp  10.10.70.2:443       192.168.0.1:443 ---             ---  
tcp  10.10.70.2:80        192.168.0.1:80 ---             ---  
Router#
```

☐ Top

Show ip nat translation on Multilayer switch since were not using router on a stick

Multilayer Switch0

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
Extended IP access list BLOCK_PRINTER_TO_STAGING
 10 deny ip 172.168.0.48 0.0.0.7 172.168.0.16 0.0.0.7
 20 permit ip any any (3 match(es))
Extended IP access list ACL_FROM_EXTERNAL_PARTNER
 10 deny ip 192.168.0.0 0.0.0.3 any
 20 permit ip any any
Extended IP access list ACL_PRINTER_TO_EXTERNAL
 10 deny ip 172.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
 20 deny ip 192.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
Extended IP access list VLAN_PRINTER_TO_FLOOR3
 10 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52
 20 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52
 30 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7 (1 match(es))
 40 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7 (1 match(es))
Extended IP access list VLAN_ISOLATION
 10 deny ip 192.168.0.0 0.0.0.7 172.168.0.32 0.0.0.7
 20 deny ip 192.168.0.0 0.0.0.7 172.168.0.40 0.0.0.7
 30 deny ip 172.168.0.32 0.0.0.7 192.168.0.0 0.0.0.7
 40 deny ip 172.168.0.40 0.0.0.7 192.168.0.0 0.0.0.7
 50 deny ip 172.168.0.32 0.0.0.7 172.168.0.40 0.0.0.7 (1 match(es))
 60 deny ip 172.168.0.40 0.0.0.7 172.168.0.32 0.0.0.7 (1 match(es))
 70 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52 (1 match(es))
 80 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52 (2 match(es))
 90 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7
100 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7
110 permit ip 172.168.0.32 0.0.0.7 172.168.0.24 0.0.0.7
120 permit ip 172.168.0.40 0.0.0.7 172.168.0.24 0.0.0.7

Switch#
Switch#
Switch#
Switch#show access-list | include hit
      ^
% Invalid input detected at '^' marker.

Switch#show access-list | include hit
Switch#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
tcp  192.168.0.1:443     172.168.0.18:443  ---                ---
tcp  192.168.0.1:80      172.168.0.18:80  ---                ---

Switch#
```

Copy

Paste

☐ Top

Show ip nat statistics

 Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>
Router>en
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
tcp  10.10.70.2:443      192.168.0.1:443   ---                ---
tcp  10.10.70.2:80        192.168.0.1:80    ---                ---

Router#show ip nat statistic
Total translations: 2 (2 static, 0 dynamic, 2 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/2
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
Router#show ip nat statistica
      ^
% Invalid input detected at '^' marker.

Router#show ip nat statisticas
      ^
% Invalid input detected at '^' marker.

Router#show ip nat statistics
Total translations: 2 (2 static, 0 dynamic, 2 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/2
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
Router#show ip nat statistics
Total translations: 2 (2 static, 0 dynamic, 2 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/2
Hits: 0 Misses: 3
Expired translations: 0
Dynamic mappings:
Router#
```

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☐ Top

Show running-config | section ip nat

 Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
tcp  10.10.70.2:443      192.168.0.1:443   ---                ---
tcp  10.10.70.2:80        192.168.0.1:80    ---                ---

Router#show ip nat statistic
Total translations: 2 (2 static, 0 dynamic, 2 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/2
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
Router#show ip nat statistica
      ^
% Invalid input detected at '^' marker.

Router#show ip nat statisticas
      ^
% Invalid input detected at '^' marker.

Router#show ip nat statistics
Total translations: 2 (2 static, 0 dynamic, 2 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/2
Hits: 0 Misses: 0
Expired translations: 0
Dynamic mappings:
Router#show ip nat statistics
Total translations: 2 (2 static, 0 dynamic, 2 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/2
Hits: 0 Misses: 3
Expired translations: 0
Dynamic mappings:
Router#show running-config | section ip nat
ip nat inside
ip nat outside
ip nat inside
ip nat inside source static tcp 192.168.0.1 80 10.10.70.2 80
ip nat inside source static tcp 192.168.0.1 443 10.10.70.2 443
Router#
```

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☐ Top

Show access list , this is for the router and the two multilayer switch , the access lists help prevent inbound and lateral movement while permitting Developers team to update the staging server without exposing additional internal resources

The image displays two Cisco IOS Command Line Interfaces (CLI) side-by-side. The left window is for 'Multilayer Switch0' and the right window is for 'Router0'. Both windows show the configuration of access lists (ACLs) for network security.

Multilayer Switch0 CLI:

```
IOS Command Line Interface
100 permit ip host 172.168.0.32 172.168.0.0/24
110 permit ip 172.168.0.32 0.0.0.7 172.168.0.24 0.0.0.7
120 permit ip 172.168.0.40 0.0.0.7 172.168.0.24 0.0.0.7

Switch#
Switch#
Switch#show access-list | include hit

% Invalid input detected at '' marker.

Switch#show access-list | include hit
Switch#show ip nat translations
Pro Inside global Inside local Outside local Outside global
top 192.168.0.11443 172.168.0.181443 --- ---
top 192.168.0.1180 172.168.0.18180 --- ---

Switch#show access-list
Extended IP access list ACL_TO_STAGING
10 permit ip 172.168.0.0 0.0.0.7 172.168.0.16 0.0.0.7
20 deny ip 172.168.0.8 0.0.0.7 172.168.0.16 0.0.0.7
30 deny ip 172.168.0.32 0.0.0.7 172.168.0.16 0.0.0.7
40 deny ip 172.168.0.40 0.0.0.7 172.168.0.16 0.0.0.7
50 permit ip any 172.168.0.16 0.0.0.7 (2 match(es))
Extended IP access list ACL_FROM_STAGING
10 deny ip 172.168.0.16 0.0.0.7 172.168.0.0 0.0.0.255
20 permit ip any any
Extended IP access list BLOCK_PRINTER_TO_STAGING
10 deny ip 172.168.0.48 0.0.0.7 172.168.0.16 0.0.0.7
20 permit ip any any (3 match(es))
Extended IP access list ACL_FROM_EXTERNAL_PARTNER
10 deny ip 192.168.0.0 0.0.0.3 any
20 permit ip any any
Extended IP access list ACL_PRINTER_TO_EXTERNAL
10 deny ip 172.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
20 deny ip 192.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
Extended IP access list VLAN_PRINTER_TO_FLOOR3
10 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52
20 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52
30 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7 (1 match(es))
40 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7 (1 match(es))
Extended IP access list VLAN_ISOLATION
10 deny ip 192.168.0.0 0.0.0.7 172.168.0.32 0.0.0.7
20 deny ip 192.168.0.0 0.0.0.7 172.168.0.40 0.0.0.7
30 deny ip 172.168.0.32 0.0.0.7 192.168.0.0 0.0.0.7
40 deny ip 172.168.0.40 0.0.0.7 192.168.0.0 0.0.0.7
50 deny ip 172.168.0.32 0.0.0.7 172.168.0.40 0.0.0.7 (1 match(es))
60 deny ip 172.168.0.40 0.0.0.7 172.168.0.32 0.0.0.7 (1 match(es))
70 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52 (1 match(es))
80 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52 (2 match(es))
90 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7
100 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7
110 permit ip 172.168.0.32 0.0.0.7 172.168.0.24 0.0.0.7
120 permit ip 172.168.0.40 0.0.0.7 172.168.0.24 0.0.0.7

Switch#
Switch#
Switch#
```

Router0 CLI:

```
IOS Command Line Interface
Router#show ip nat statistics
Total translations: 2 (2 static, 0 dynamic, 2 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0, GigabitEthernet0/2
Hits: 0 Misses: 3
Expired translations: 0
Dynamic mappings:
Router#show running-config | section ip nat
ip nat outside
ip nat inside
ip nat inside source static top 192.168.0.1 80 10.10.20.2 80
ip nat inside source static top 192.168.0.1 443 10.10.20.2 443
Router#
Router#show access-list
Extended IP access list VLAN_ISOLATION
10 deny ip 192.168.0.0 0.0.0.3 172.168.0.32 0.0.0.7
20 deny ip 192.168.0.0 0.0.0.3 172.168.0.40 0.0.0.7
30 permit top 192.168.0.0 0.0.0.3 host 192.168.0.1 eq www
40 permit top 192.168.0.0 0.0.0.3 host 192.168.0.1 eq 443
50 deny ip any any (979 match(es))
Router#
```

Multilayer Switch1 CLI:

```
IOS Command Line Interface
Switch#en
Switch#show access-list

% Invalid input detected at '' marker.

Switch#show access-list

% Invalid input detected at '' marker.

Switch#show access-list
Extended IP access list BLOCK_EXTERNAL
10 deny ip 192.168.1.0 0.0.0.7 10.10.20.0 0.0.0.3
20 deny ip 192.168.1.0 0.0.0.7 172.168.0.24 0.0.0.7
30 deny ip 192.168.1.0 0.0.0.7 10.10.40.0 0.0.0.3
40 deny ip 192.168.1.0 0.0.0.7 172.168.0.0 0.0.0.7
50 deny ip 192.168.1.0 0.0.0.7 172.168.0.8 0.0.0.7
60 permit ip 10.10.40.0 0.0.0.3 172.168.0.24 0.0.0.7
70 permit ip 10.10.20.0 0.0.0.3 172.168.0.24 0.0.0.7
80 permit ip 172.168.0.24 0.0.0.7 10.10.40.0 0.0.0.3
90 permit ip 172.168.0.24 0.0.0.7 10.10.20.0 0.0.0.3
100 permit ip any any (947 match(es))

Switch#
```


Hit Count

Physical Config CLI Attributes

IOS Command Line Interface

```
40 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7 (1 match(es))
Extended IP access list VLAN_ISOLATION
10 deny ip 192.168.0.0 0.0.0.7 172.168.0.32 0.0.0.7
20 deny ip 192.168.0.0 0.0.0.7 172.168.0.40 0.0.0.7
30 deny ip 172.168.0.32 0.0.0.7 192.168.0.0 0.0.0.7
40 deny ip 172.168.0.40 0.0.0.7 192.168.0.0 0.0.0.7
50 deny ip 172.168.0.32 0.0.0.7 172.168.0.40 0.0.0.7 (1 match(es))
60 deny ip 172.168.0.40 0.0.0.7 172.168.0.32 0.0.0.7 (1 match(es))
70 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52 (1 match(es))
80 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52 (2 match(es))
90 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7
100 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7
110 permit ip 172.168.0.32 0.0.0.7 172.168.0.24 0.0.0.7
120 permit ip 172.168.0.40 0.0.0.7 172.168.0.24 0.0.0.7

Switch#
Switch#
Switch#show access-list
Extended IP access list ACL_TO_STAGING
10 permit ip 172.168.0.0 0.0.0.7 172.168.0.16 0.0.0.7
20 deny ip 172.168.0.8 0.0.0.7 172.168.0.16 0.0.0.7
30 deny ip 172.168.0.32 0.0.0.7 172.168.0.16 0.0.0.7
40 deny ip 172.168.0.40 0.0.0.7 172.168.0.16 0.0.0.7
50 permit ip any 172.168.0.16 0.0.0.7 (8 match(es))
Extended IP access list ACL_FROM_STAGING
10 deny ip 172.168.0.16 0.0.0.7 172.168.0.0 0.0.0.255
20 permit ip any any
Extended IP access list BLOCK_PRINTER_TO_STAGING
10 deny ip 172.168.0.48 0.0.0.7 172.168.0.16 0.0.0.7
20 permit ip any any (3 match(es))
Extended IP access list ACL_FROM_EXTERNAL_PARTNER
10 deny ip 192.168.0.0 0.0.0.3 any
20 permit ip any any
Extended IP access list ACL_PRINTER_TO_EXTERNAL
10 deny ip 172.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
20 deny ip 192.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7
Extended IP access list VLAN_PRINTER_TO_FLOOR3
10 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52
20 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52
30 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7 (1 match(es))
40 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7 (1 match(es))
Extended IP access list VLAN_ISOLATION
10 deny ip 192.168.0.0 0.0.0.7 172.168.0.32 0.0.0.7
20 deny ip 192.168.0.0 0.0.0.7 172.168.0.40 0.0.0.7
30 deny ip 172.168.0.32 0.0.0.7 192.168.0.0 0.0.0.7
40 deny ip 172.168.0.40 0.0.0.7 192.168.0.0 0.0.0.7
50 deny ip 172.168.0.32 0.0.0.7 172.168.0.40 0.0.0.7 (1 match(es))
60 deny ip 172.168.0.40 0.0.0.7 172.168.0.32 0.0.0.7 (1 match(es))
70 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52 (1 match(es))
80 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52 (2 match(es))
90 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7
100 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7
110 permit ip 172.168.0.32 0.0.0.7 172.168.0.24 0.0.0.7 (2 match(es))
120 permit ip 172.168.0.40 0.0.0.7 172.168.0.24 0.0.0.7 (1 match(es))

Switch#
Switch#
Switch#
```

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IOS Command Line Interface

```
20 deny ip 192.168.1.0 0.0.0.7 10.10.20.0 0.0.0.3
20 deny ip 192.168.1.0 0.0.0.7 172.168.0.24 0.0.0.7
30 deny ip 192.168.1.0 0.0.0.7 10.10.40.0 0.0.0.3
40 deny ip 192.168.1.0 0.0.0.7 172.168.0.0 0.0.0.7
50 deny ip 192.168.1.0 0.0.0.7 172.168.0.8 0.0.0.7
60 permit ip 10.10.40.0 0.0.0.3 172.168.0.24 0.0.0.7
70 permit ip 10.10.20.0 0.0.0.3 172.168.0.24 0.0.0.7
80 permit ip 172.168.0.24 0.0.0.7 10.10.40.0 0.0.0.3
90 permit ip 172.168.0.24 0.0.0.7 10.10.20.0 0.0.0.3
100 permit ip any any (947 match(es))

Switch#show access-list
Extended IP access list BLOCK_EXTERNAL
10 deny ip 192.168.1.0 0.0.0.7 10.10.20.0 0.0.0.3
20 deny ip 192.168.1.0 0.0.0.7 172.168.0.24 0.0.0.7
30 deny ip 192.168.1.0 0.0.0.7 10.10.40.0 0.0.0.3
40 deny ip 192.168.1.0 0.0.0.7 172.168.0.0 0.0.0.7 (1 match(es))
50 deny ip 192.168.1.0 0.0.0.7 172.168.0.8 0.0.0.7 (1 match(es))
60 permit ip 10.10.40.0 0.0.0.3 172.168.0.24 0.0.0.7
70 permit ip 10.10.20.0 0.0.0.3 172.168.0.24 0.0.0.7
80 permit ip 172.168.0.24 0.0.0.7 10.10.40.0 0.0.0.3
90 permit ip 172.168.0.24 0.0.0.7 10.10.20.0 0.0.0.3
100 permit ip any any (998 match(es))

Switch#
```

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Router0

IOS Command Line Interface

```
ip nat inside
ip nat inside source static tcp 192.168.0.1 80 10.10.70.2 80
ip nat inside source static tcp 192.168.0.1 443 10.10.70.2 443

Router#
Router#
Router#show access-list
Extended IP access list VLAN-ISOLATION
Extended IP access list ACL_OUTSIDE_IN
10 deny ip 192.168.0.0 0.0.0.3 172.168.0.32 0.0.0.7
20 deny ip 192.168.0.0 0.0.0.3 172.168.0.40 0.0.0.7
30 permit tcp 192.168.0.0 0.0.0.3 host 192.168.0.1 eq www
40 permit tcp 192.168.0.0 0.0.0.3 host 192.168.0.1 eq 443
50 deny ip any any (879 match(es))

Router#show access-list
Extended IP access list VLAN-ISOLATION
Extended IP access list ACL_OUTSIDE_IN
10 deny ip 192.168.0.0 0.0.0.3 172.168.0.32 0.0.0.7
20 deny ip 192.168.0.0 0.0.0.3 172.168.0.40 0.0.0.7
30 permit tcp 192.168.0.0 0.0.0.3 host 192.168.0.1 eq www
40 permit tcp 192.168.0.0 0.0.0.3 host 192.168.0.1 eq 443
50 deny ip any any (939 match(es))

Router#
```

Copy Paste

Show logging

Physical

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IOS Command Line Interface

10 deny ip 172.168.0.48 0.0.0.7 172.168.0.48 0.0.0.7

20 permit ip any any (3 match(es))

Extended IP access list ACL_FROM_EXTERNAL_PARTNER

10 deny ip 192.168.0.0 0.0.0.3 any

20 permit ip any any

Extended IP access list ACL_PRINTER_TO_EXTERNAL

10 deny ip 172.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7

20 deny ip 192.168.0.48 0.0.0.7 192.168.0.0 0.0.0.7

Extended IP access list TEAM_PRINTER_TO_FLOORS

10 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52

20 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52

30 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7 (1 match(es))

40 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7 (1 match(es))

Extended IP access list TEAM_ISOLATION

10 deny ip 192.168.0.0 0.0.0.7 172.168.0.32 0.0.0.7

20 deny ip 192.168.0.0 0.0.0.7 172.168.0.40 0.0.0.7

30 deny ip 172.168.0.32 0.0.0.7 192.168.0.0 0.0.0.7

40 deny ip 172.168.0.40 0.0.0.7 192.168.0.0 0.0.0.7

50 deny ip 172.168.0.32 0.0.0.7 172.168.0.40 0.0.0.7 (1 match(es))

60 deny ip 172.168.0.40 0.0.0.7 172.168.0.32 0.0.0.7 (1 match(es))

70 permit ip 172.168.0.32 0.0.0.7 host 172.168.0.52 (1 match(es))

80 permit ip 172.168.0.40 0.0.0.7 host 172.168.0.52 (2 match(es))

90 permit ip host 172.168.0.52 172.168.0.32 0.0.0.7

100 permit ip host 172.168.0.52 172.168.0.40 0.0.0.7

110 permit ip 172.168.0.32 0.0.0.7 172.168.0.24 0.0.0.7 (2 match(es))

120 permit ip 172.168.0.40 0.0.0.7 172.168.0.24 0.0.0.7 (1 match(es))

Switch#

Switch#

Switch#show logging

Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 25 messages logged, xml disabled,
filtering disabled

Monitor logging: level debugging, 25 messages logged, xml disabled,
filtering disabled

Buffer logging: disabled, xml disabled,
filtering disabled

Logging Exception size (4096 bytes)

Count and timestamp logging messages: disabled

Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped

Trap logging: level informational, 25 message lines logged

Switch#

Switch#

Switch#

Switch#

Copy

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IOS Command Line Interface

Switch#show logging

Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 13 messages logged, xml disabled,
filtering disabled

Monitor logging: level debugging, 13 messages logged, xml disabled,
filtering disabled

Buffer logging: disabled, xml disabled,
filtering disabled

Logging Exception size (4096 bytes)

Count and timestamp logging messages: disabled

Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped

Trap logging: level informational, 13 message lines logged

Copy

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Router0

IOS Command Line Interface

Router#show logging

Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 5 messages logged, xml disabled,
filtering disabled

Monitor logging: level debugging, 5 messages logged, xml disabled,
filtering disabled

Buffer logging: disabled, xml disabled,
filtering disabled

Logging Exception size (4096 bytes)

Count and timestamp logging messages: disabled

Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped

Trap logging: level informational, 5 message lines logged

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Top

3.2) In the topology, the staging server is isolated via VLANs on the multilayer switch.

By placing the Staging Server on a dedicated VLAN on the Multilayer switch, we are isolating it and keeping the server secure. It also helps creating a controlled buffer. This ensures that external partners accessing the staging site do not interact with the internal corporate LAN. This physical and logical separation limits "east-west" movement, meaning if the Staging Server is compromised, the attacker cannot easily hop to the sensitive Developer or BA networks on Floor 1 and Test engineers and UI designers who are on floor 2.

Data Plane Policies will be applied on the two multilayer switches to manage traffic flow and restrict lateral movement:

All other subnets will be /have been denied access to the staging server besides Developers and external partners. External partners will only have the ability to access HTTP(S) services while developers will be able to manage the server.

The routing protocol to be used is OSPF. This is because it provides routing protocol authentication which helps prevent rogue route injection.

An Access Control List (ACL) on the Shared Router's WAN interface and the Floor 2 and 3 Multi layer switch will allow HTTP/HTTPS (TCP 80/443) traffic from specific Partner IP addresses to the Staging Server's public NAT IP. A specific permit rule will allow the Floor 1 Developer to access the staging server and an explicit "Deny" rule must prevent the Staging Server (Floor 3) from initiating connections back into Floor 1 or Floor 2. On the rest of the network more ACLs will be deployed to prevent external partners from accessing any other internal subnets besides the staging server.

We will adopt a Split Horizon DNS method.

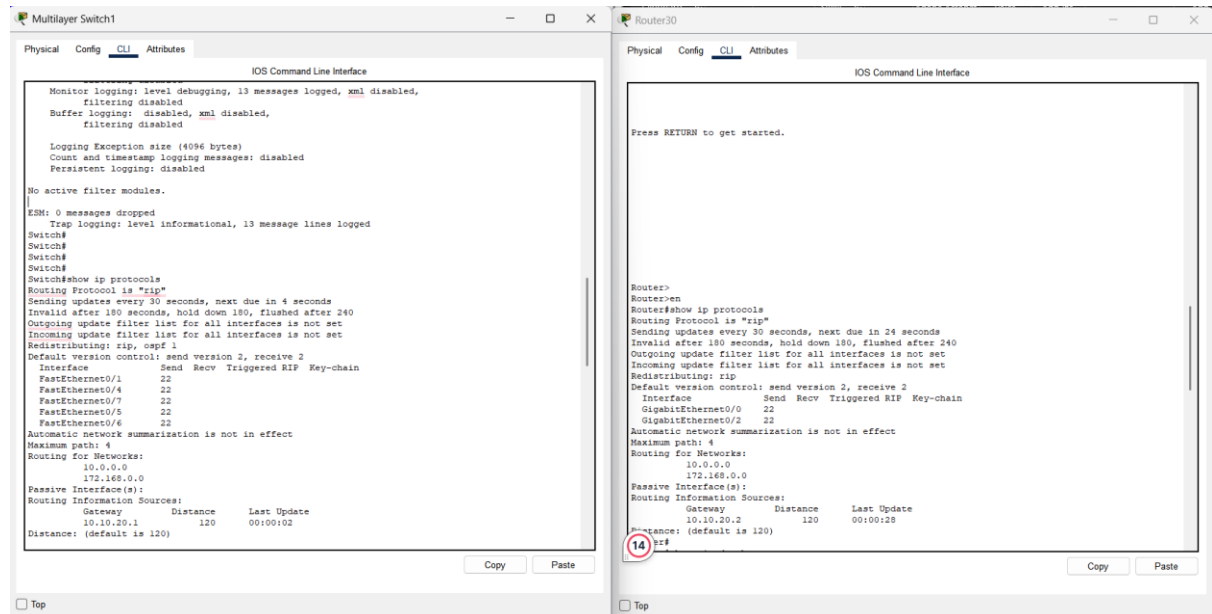
Internal: When Developers on Floor 1 query staging.brightsoft.com, the internal DNS returns the

server's private, local IP address (e.g., 172.168.0.18)

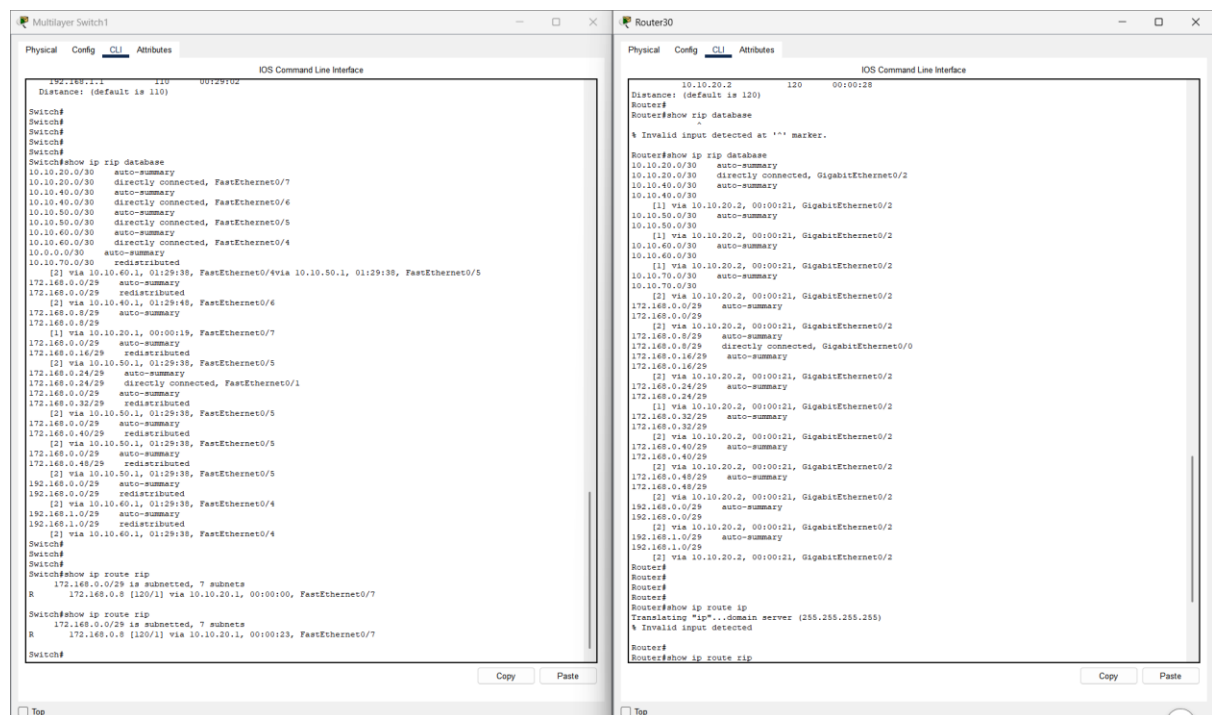
External: When partners query the same URL, a public DNS returns the public NAT address. This hides the internal network structure from the outside world.

Question 4

4.1) Show ip protocols on business analyst router and the multilayer switch that connects all three floors



Show RIP database



Show IP route RIP

Physical Config CLI Attributes

IOS Command Line Interface

```
192.168.1.1 110 00129102
Distance: (default is 110)

Switch#
Switch#
Switch#
Switch#
Switch#show ip rip database
10.10.20.0/30 auto-summary
10.10.20.0/30 directly connected, FastEthernet0/7
10.10.40.0/30 auto-summary
10.10.40.0/30 directly connected, FastEthernet0/6
10.10.50.0/30 auto-summary
10.10.50.0/30 directly connected, FastEthernet0/5
10.10.60.0/30 auto-summary
10.10.60.0/30 directly connected, FastEthernet0/4
10.0.0.0/30 auto-summary
10.10.70.0/30 redistributed
[2] via 10.10.60.1, 01:29:38, FastEthernet0/4 via 10.10.50.1, 01:29:38, FastEthernet0/5
172.168.0.0/29 auto-summary
172.168.0.0/29 redistributed
[2] via 10.10.40.1, 01:29:48, FastEthernet0/6
172.168.0.8/29 auto-summary
172.168.0.8/29
[1] via 10.10.20.1, 00:00:19, FastEthernet0/7
172.168.0.0/29 auto-summary
172.168.0.16/29 redistributed
[2] via 10.10.50.1, 01:29:38, FastEthernet0/5
172.168.0.24/29 auto-summary
172.168.0.24/29 directly connected, FastEthernet0/1
172.168.0.0/29 auto-summary
172.168.0.32/29 redistributed
[2] via 10.10.50.1, 01:29:38, FastEthernet0/5
172.168.0.0/29 auto-summary
172.168.0.40/29 redistributed
[2] via 10.10.50.1, 01:29:38, FastEthernet0/5
172.168.0.0/29 auto-summary
172.168.0.48/29 redistributed
[2] via 10.10.50.1, 01:29:38, FastEthernet0/5
192.168.0.0/29 auto-summary
192.168.0.0/29 redistributed
[2] via 10.10.40.1, 01:29:38, FastEthernet0/4
192.168.1.0/29 auto-summary
192.168.1.0/29 redistributed
[2] via 10.10.40.1, 01:29:38, FastEthernet0/4
Switch#
Switch#
Switch#
Switch#show ip route rip
172.168.0.0/29 is subnetted, 7 subnets
R 172.168.0.8 [120/2] via 10.10.20.1, 00:00:00, FastEthernet0/7
Switch#show ip route rip
172.168.0.0/29 is subnetted, 7 subnets
R 172.168.0.8 [120/1] via 10.10.20.1, 00:00:23, FastEthernet0/7
Switch#
```

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Physical Config CLI Attributes

IOS Command Line Interface

```
[2] via 10.10.20.1, 00:00:21, GigabitEthernet0/2
10.10.70.0/30 auto-summary
10.10.70.0/30
[2] via 10.10.20.2, 00:00:21, GigabitEthernet0/2
172.168.0.0/29 auto-summary
172.168.0.0/29
[2] via 10.10.20.2, 00:00:21, GigabitEthernet0/2
172.168.0.8/29 auto-summary
172.168.0.8/29 directly connected, GigabitEthernet0/0
172.168.0.16/29 auto-summary
172.168.0.16/29
[2] via 10.10.20.2, 00:00:21, GigabitEthernet0/2
172.168.0.24/29 auto-summary
172.168.0.24/29
[1] via 10.10.20.2, 00:00:21, GigabitEthernet0/2
172.168.0.32/29 auto-summary
172.168.0.32/29
[2] via 10.10.20.2, 00:00:21, GigabitEthernet0/2
172.168.0.40/29 auto-summary
172.168.0.40/29
[2] via 10.10.20.2, 00:00:21, GigabitEthernet0/2
172.168.0.48/29 auto-summary
172.168.0.48/29
[2] via 10.10.20.2, 00:00:21, GigabitEthernet0/2
192.168.0.0/29 auto-summary
192.168.0.0/29
[2] via 10.10.20.2, 00:00:21, GigabitEthernet0/2
192.168.1.0/29 auto-summary
192.168.1.0/29
[2] via 10.10.20.2, 00:00:21, GigabitEthernet0/2
Router#
Router#
Router#
Router#show ip route ip
Translating 'ip'...domain server (255.255.255.255)
% Invalid input detected
Router#
Router#show ip route rip
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
R 10.10.40.0/30 [120/1] via 10.10.20.2, 00:00:08, GigabitEthernet0/2
R 10.10.50.0/30 [120/1] via 10.10.20.2, 00:00:08, GigabitEthernet0/2
R 10.10.60.0/30 [120/1] via 10.10.20.2, 00:00:08, GigabitEthernet0/2
R 10.10.70.0/30 [120/2] via 10.10.20.2, 00:00:08, GigabitEthernet0/2
172.168.0.0/16 is variably subnetted, 8 subnets, 2 masks
R 172.168.0.0/29 [120/2] via 10.10.20.2, 00:00:08, GigabitEthernet0/2
R 172.168.0.16/29 [120/2] via 10.10.20.2, 00:00:08, GigabitEthernet0/2
R 172.168.0.32/29 [120/2] via 10.10.20.2, 00:00:08, GigabitEthernet0/2
R 172.168.0.40/29 [120/2] via 10.10.20.2, 00:00:08, GigabitEthernet0/2
R 172.168.0.48/29 [120/2] via 10.10.20.2, 00:00:08, GigabitEthernet0/2
192.168.0.0/29 is subnetted, 1 subnets
R 192.168.0.0 [120/2] via 10.10.20.2, 00:00:08, GigabitEthernet0/2
192.168.1.0/29 is subnetted, 1 subnets
R 192.168.1.0 [120/2] via 10.10.20.2, 00:00:08, GigabitEthernet0/2
Router#
```

Copy Paste

Top

Show ip ospf neighbour on the router and two multilayer switches

The image displays the Cisco Packet Tracer interface with three windows open, showing the output of the 'show ip ospf neighbor' command on different devices.

Router0 CLI:

```
Router>
Router>en
Router#show ip ospf neighbor
% Invalid input detected at '' marker.

Router#show ip ospf neighbor

Neighbor ID      Pri  State      Dead Time   Address      Interface
192.168.1.1      1    FULL/DR    00:00:39    192.168.0.2  GigabitEthernet0/1
1.1.1.1          1    FULL/DR    00:00:39    10.10.60.2   GigabitEthernet0/2

Router#show ip ospf interface

GigabitEthernet0/2 is up, line protocol is up
Internet address is 10.10.60.1/30, Area 0
Process ID 1, Router ID 192.168.0.1, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 192.168.0.1, Interface address 10.10.60.1
Backup Designated Router (ID) 1.1.1.1, Interface address 10.10.60.2
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:06
Index 1/1, flood queue length 0
#show ip ospf neighbors
```

Multilayer Switch0 CLI:

```
Switch#
Switch#en
Switch#show ip ospf neighbor
% Invalid input detected at '' marker.

Switch#show ip ospf neighbor

Neighbor ID      Pri  State      Dead Time   Address      Interface
192.168.0.1      1    INIT/BROTHER 00:00:39    10.10.70.1   FastEthernet0/2
1.1.1.1          1    FULL/DR    00:00:36    10.10.50.2   FastEthernet0/5

Switch#
```

Multilayer Switch1 CLI:

```
172.168.0.0/29 auto-summary
172.168.0.0/29
[1] via 10.10.20.1, 00:00:19, FastEthernet0/7
172.168.0.0/29 auto-summary
172.168.0.16/29 redistributed
[2] via 10.10.50.1, 01:29:39, FastEthernet0/5
172.168.0.0/29 auto-summary
172.168.0.24/29 auto-summary
172.168.0.24/29 directly connected, FastEthernet0/1
172.168.0.0/29 auto-summary
172.168.0.32/29 redistributed
[2] via 10.10.50.1, 01:29:39, FastEthernet0/5
172.168.0.0/29 auto-summary
172.168.0.40/29 redistributed
[2] via 10.10.50.1, 01:29:39, FastEthernet0/5
172.168.0.0/29 auto-summary
172.168.0.48/29 redistributed
[2] via 10.10.50.1, 01:29:39, FastEthernet0/5
192.168.0.0/29 auto-summary
192.168.0.0/29 redistributed
[2] via 10.10.60.1, 01:29:39, FastEthernet0/4
192.168.1.0/29 auto-summary
192.168.1.0/29 redistributed
[2] via 10.10.60.1, 01:29:39, FastEthernet0/4

Switch#
Switch#
Switch#show ip route rip
172.168.0.0/29 is subnetted, 7 subnets
R 172.168.0.8 [120/1] via 10.10.20.1, 00:00:00, FastEthernet0/7

Switch#show ip route rip
172.168.0.0/29 is subnetted, 7 subnets
R 172.168.0.8 [120/1] via 10.10.20.1, 00:00:23, FastEthernet0/7

Switch#show ip ospf neighbor

Neighbor ID      Pri  State      Dead Time   Address      Interface
192.168.0.1      1    FULL/DR    00:00:38    10.10.60.1   FastEthernet0/4
172.168.0.1      1    FULL/DR    00:00:37    10.10.40.1   FastEthernet0/4
172.168.0.51      1    FULL/DR    00:00:37    10.10.50.1   FastEthernet0/5

Switch#
```

Show ip ospf interface

Multilayer switch serving as a bridge

```
IOS Command Line Interface

FastEthernet0/1 is up, line protocol is up
Internet address is 172.168.0.25/25, Area 0
Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 1.1.1.1, Interface address 172.168.0.25
No backup designated router on this network
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:03
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 0, Adjacent neighbor count is 0
Suppress hello for 0 neighbor(s)
FastEthernet0/4 is up, line protocol is up
Internet address is 10.10.60.2/30, Area 0
Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State BDR, Priority 1
Designated Router (ID) 192.168.0.1, Interface address 10.10.60.1
Backup Designated Router (ID) 1.1.1.1, Interface address 10.10.60.2
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:03
Index 2/2, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 192.168.0.1 (Designated Router)
Suppress hello for 0 neighbor(s)
FastEthernet0/5 is up, line protocol is up
Internet address is 10.10.60.2/30, Area 0
Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State BDR, Priority 1
Designated Router (ID) 172.168.0.51, Interface address 10.10.50.1
Backup Designated Router (ID) 1.1.1.1, Interface address 10.10.50.2
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:03
Index 3/3, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 172.168.0.51 (Designated Router)
Suppress hello for 0 neighbor(s)
FastEthernet0/6 is up, line protocol is up
Internet address is 10.10.40.2/30, Area 0
Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State BDR, Priority 1
Designated Router (ID) 172.168.0.1, Interface address 10.10.40.1
Backup Designated Router (ID) 1.1.1.1, Interface address 10.10.40.2
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:02
Index 4/4, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 172.168.0.1 (Designated Router)
Suppress hello for 0 neighbor(s)
```

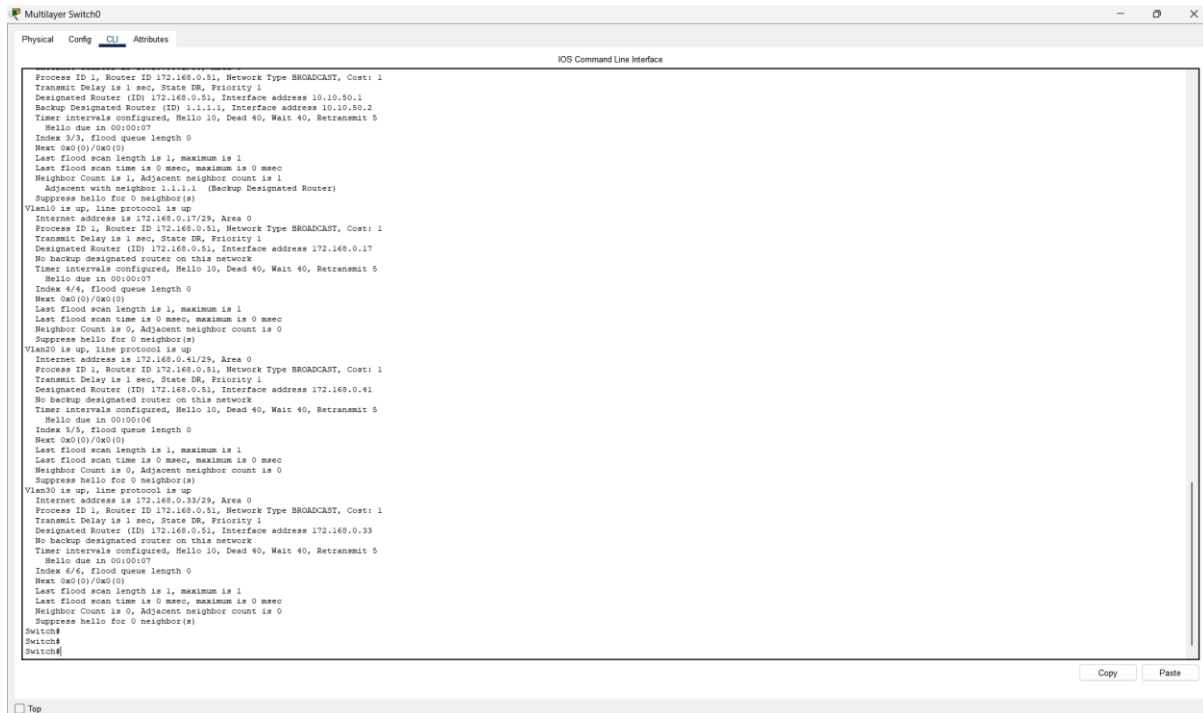
Multilayer switch for Floor 2 and 3

Part 1



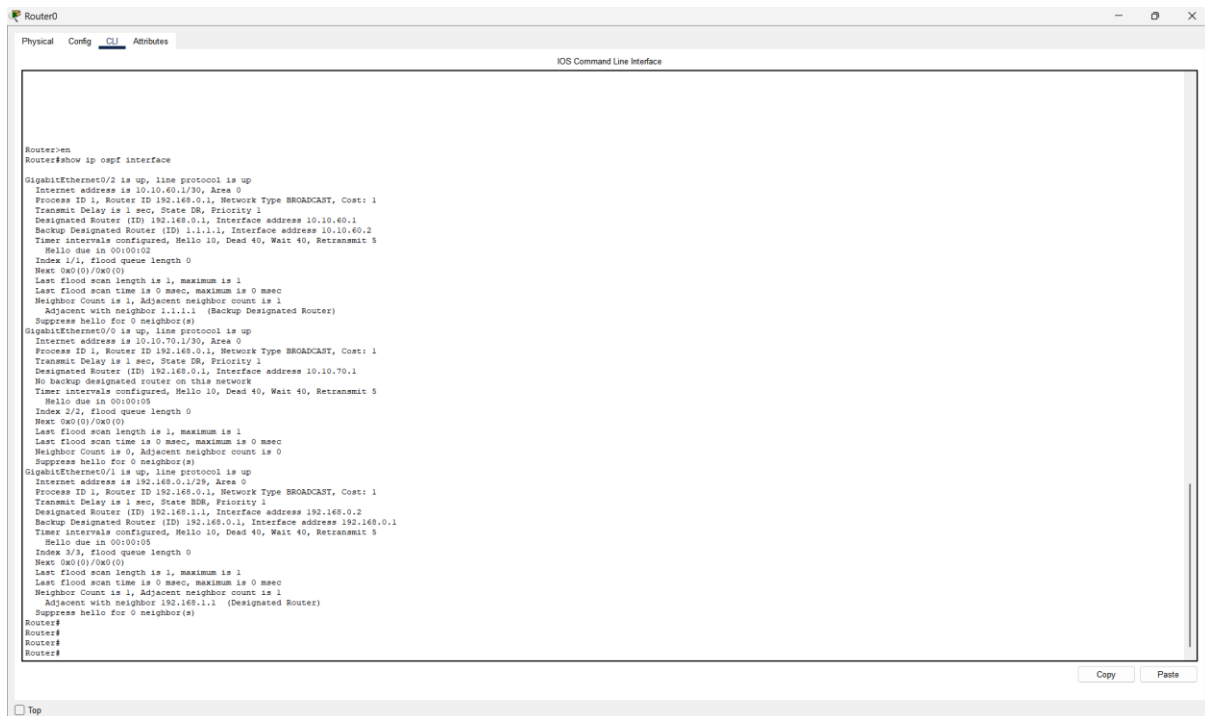
```
Switch0#show ip ospf interface
FastEthernet0/24 is up, line protocol is up
  Internet address is 172.168.0.51/29, Area 0
  Process ID 1, Router ID 172.168.0.51, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 172.168.0.51, Interface address 172.168.0.51
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:07
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
FastEthernet0/25 is up, line protocol is up
  Internet address is 10.10.70.2/30, Area 0
  Process ID 1, Router ID 172.168.0.51, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 172.168.0.51, Interface address 10.10.70.2
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:07
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 192.168.0.1
  Suppress hello for 0 neighbor(s)
FastEthernet0/26 is up, line protocol is up
  Internet address is 10.10.50.1/30, Area 0
  Process ID 1, Router ID 172.168.0.51, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 172.168.0.51, Interface address 10.10.50.1
  Backup Designated Router (ID) 1.1.1.1, Interface address 10.10.50.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:07
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 1.1.1.1 (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
Vlan0 is up, line protocol is up
  Internet address is 172.168.0.17/29, Area 0
  Process ID 1, Router ID 172.168.0.51, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 172.168.0.51, Interface address 172.168.0.17
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:07
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
```

Part 2

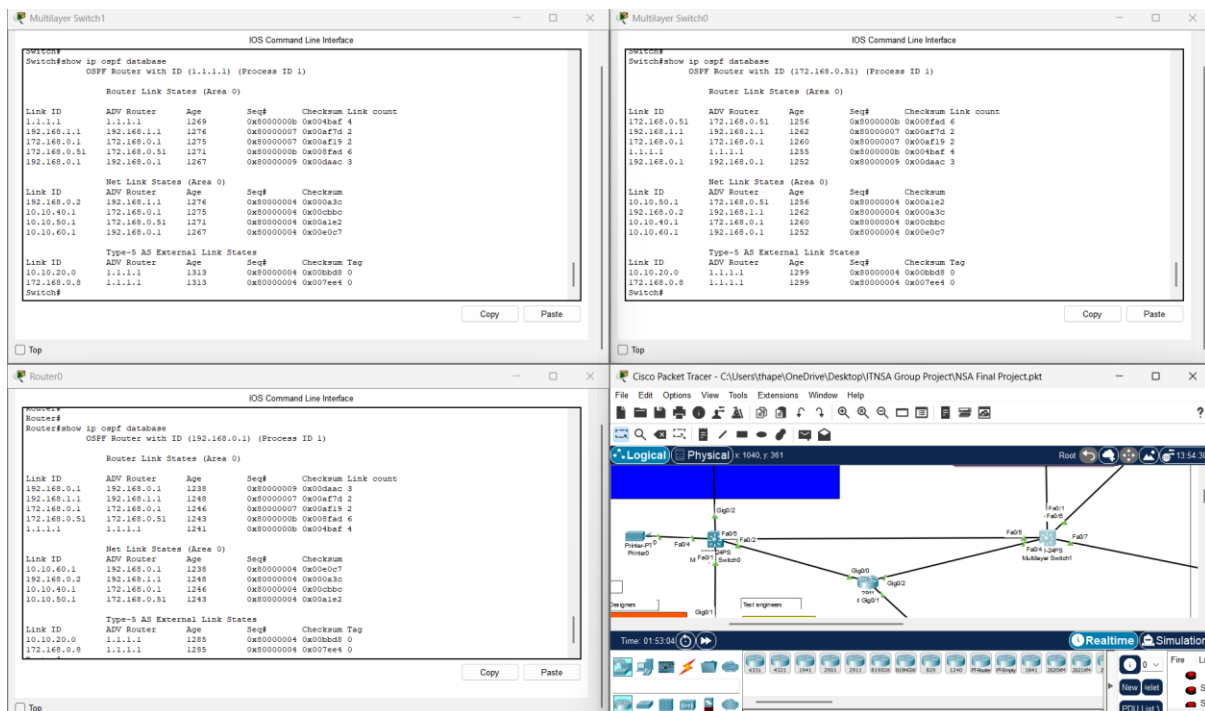


```
Process ID 1, Router ID 172.168.0.51, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 172.168.0.51, Interface address 10.10.50.1
  Backup Designated Router (ID) 1.1.1.1, Interface address 10.10.50.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:07
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 1.1.1.1 (Backup Designated Router)
  Suppress hello for 0 neighbor(s)
Vlan0 is up, line protocol is up
  Internet address is 172.168.0.17/29, Area 0
  Process ID 1, Router ID 172.168.0.51, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 172.168.0.51, Interface address 172.168.0.17
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:07
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Vlan0 is up, line protocol is up
  Internet address is 172.168.0.41/29, Area 0
  Process ID 1, Router ID 172.168.0.51, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 172.168.0.51, Interface address 172.168.0.41
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:06
  Index 5/5, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Vlan0 is up, line protocol is up
  Internet address is 172.168.0.33/29, Area 0
  Process ID 1, Router ID 172.168.0.51, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 172.168.0.51, Interface address 172.168.0.33
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:07
  Index 6/6, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Switch#
Switch#
```


Show ip ospf interface on the shared router



Show ip ospf database for all 3 Layer 3 devices



Show ip route OSPF for all three layer 3 devices

The image displays three screenshots of Cisco IOS Command Line Interface (CLI) sessions, showing the output of the `show ip route ospf` command for three different layer 3 devices: a switch and two routers.

Switch#

```
IOS Command Line Interface
192.168.0.2 192.168.1.1 1276 0x80000004 0x00a3c
10.10.40.1 172.168.0.1 1275 0x80000004 0x00abb
10.10.50.1 172.168.0.51 1271 0x80000004 0x00a1e2
10.10.60.1 192.168.0.1 1267 0x80000004 0x00e0c7

Type-5 AS External Link States
Link ID ADV Router Age Seq# Checksum Tag
10.10.20.0 1.1.1.1 1313 0x80000004 0x00bbd5 0
172.168.0.8 1.1.1.1 1313 0x80000004 0x007ee4 0

Switch#show ip route ospf
10.0.0.0/30 is subnetted, 5 subnets
O 10.10.70.0 [110/2] via 10.10.60.1, 01:53:15, FastEthernet0/4
  (110/2) via 10.10.50.1, 01:53:15, FastEthernet0/5
172.168.0.0/29 is subnetted, 7 subnets
O 172.168.0.0 [110/2] via 10.10.40.1, 01:53:15, FastEthernet0/6
O 172.168.0.16 [110/2] via 10.10.50.1, 01:53:15, FastEthernet0/5
O 172.168.0.32 [110/2] via 10.10.50.1, 01:53:15, FastEthernet0/5
O 172.168.0.48 [110/2] via 10.10.50.1, 01:53:15, FastEthernet0/5
O 172.168.0.64 [110/2] via 10.10.50.1, 01:53:15, FastEthernet0/5
O 192.168.0.0/29 is subnetted, 1 subnets
O 192.168.0.0 [110/2] via 10.10.40.1, 01:53:15, FastEthernet0/4
O 192.168.1.0/29 is subnetted, 1 subnets
O 192.168.1.0 [110/3] via 10.10.60.1, 01:53:15, FastEthernet0/4

Switch#
```

Router0

```
IOS Command Line Interface
192.168.0.2 192.168.1.1 1248 0x80000004 0x00a3c
10.10.40.1 172.168.0.1 1246 0x80000004 0x00abb
10.10.50.1 172.168.0.51 1243 0x80000004 0x00a1e2

Type-5 AS External Link States
Link ID ADV Router Age Seq# Checksum Tag
10.10.20.0 1.1.1.1 1285 0x80000004 0x00bbd5 0
172.168.0.8 1.1.1.1 1285 0x80000004 0x007ee4 0

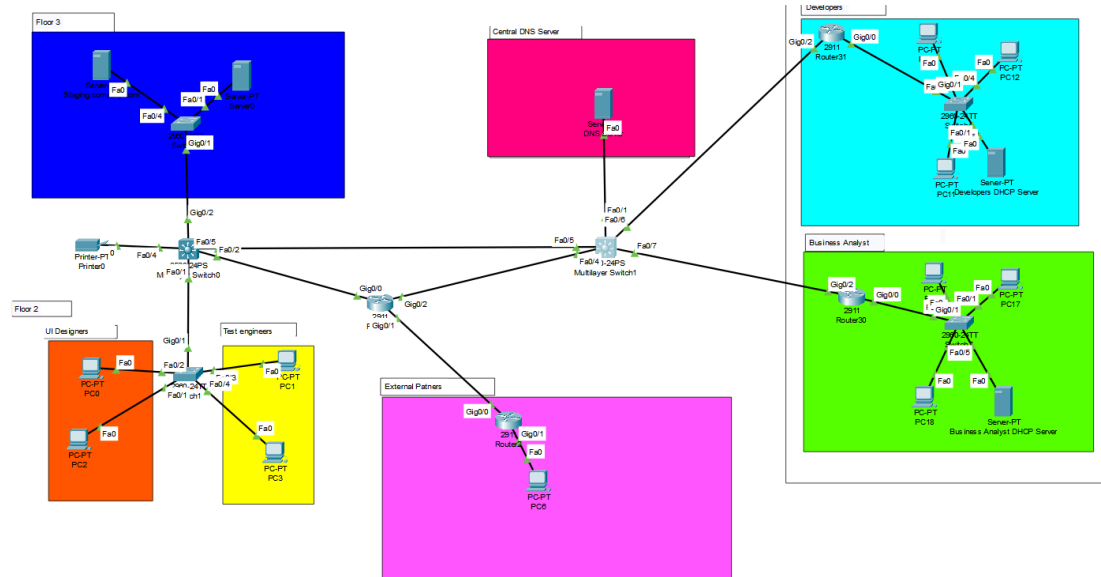
Router#
Router#show ip route ospf
10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O E2 10.10.20.0 [110/20] via 10.10.60.2, 01:53:24, GigabitEthernet0/2
O 10.10.40.0 [110/2] via 10.10.60.2, 01:53:24, GigabitEthernet0/2
O 10.10.50.0 [110/2] via 10.10.60.2, 01:53:24, GigabitEthernet0/2
172.168.0.0/29 is subnetted, 7 subnets
O 172.168.0.0 [110/3] via 10.10.60.2, 01:53:24, GigabitEthernet0/2
O E2 172.168.0.8 [110/20] via 10.10.40.2, 01:53:24, GigabitEthernet0/2
O 172.168.0.24 [110/2] via 10.10.60.2, 01:53:24, GigabitEthernet0/2
O 172.168.0.32 [110/3] via 10.10.60.2, 01:53:24, GigabitEthernet0/2
O 172.168.0.40 [110/3] via 10.10.60.2, 01:53:24, GigabitEthernet0/2
O 172.168.0.48 [110/3] via 10.10.60.2, 01:53:24, GigabitEthernet0/2
O 192.168.1.0/29 is subnetted, 1 subnets
O 192.168.1.0 [110/2] via 192.168.0.2, 01:53:34, GigabitEthernet0/1

Router#
```

Q4.1 - RIP, OSPF, RIPv2 Configuration

Question	Area	Required Verification Commands / Evidence
OSPF verification		show ip ospf neighbor
		show ip ospf interface
		show ip ospf database
		show ip route ospf
Backup link evidence		show interfaces description
show ip route		
Q4.2 - Theory	N/A	No commands required

Backup link is the straight connection from the two multilayer switch (fa/5 -fa0/5)



Show ip route for the Shared router and the Floor 2 and 3 multi-layer switch

Physical

Config

CLI

Attributes

IOS Command Line Interface

```

10.10.50.1 172.168.0.51 1256 0x00000004 0x00000000
192.168.0.2 192.168.1.1 1262 0x00000004 0x00000000
10.10.40.1 172.168.0.1 1260 0x00000004 0x00000000
10.10.40.1 192.168.0.1 1252 0x00000004 0x00000000

Type-5 AD External Link States
Link ID ADV Router Age Seq# Checksum Tag
10.10.20.0 1.1.1.1 1299 0x00000004 0x00000000 0
172.168.0.8 1.1.1.1 1299 0x00000004 0x00000000 0

Switch#show ip route ospf
10.0.0.0/30 is subnetted, 8 subnets
O E2 10.10.20.0 [110/20] via 10.10.50.2, 01:53:12, FastEthernet0/5
O 10.10.40.0 [110/2] via 10.10.50.2, 01:53:12, FastEthernet0/5
O 10.10.40.0 [110/2] via 10.10.50.2, 01:53:12, FastEthernet0/5
O 172.168.0.0/29 is subnetted, 7 subnets
O 172.168.0.0 [110/3] via 10.10.50.2, 01:53:12, FastEthernet0/5
O E2 172.168.0.8 [110/20] via 10.10.50.2, 01:53:12, FastEthernet0/5
O 172.168.0.24 [110/2] via 10.10.50.2, 01:53:12, FastEthernet0/5
O E2 172.168.0.24 [110/20] via 10.10.50.2, 01:53:12, FastEthernet0/5
O 192.168.0.0/29 is subnetted, 1 subnets
O 192.168.0.0 [110/3] via 10.10.50.2, 01:53:02, FastEthernet0/5
O 192.168.1.0/29 is subnetted, 1 subnets
O 192.168.1.0 [110/4] via 10.10.50.2, 01:53:02, FastEthernet0/5

Switch#show ip route
Codes: C - connected, S - static, I - IGMP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
NI - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/30 is subnetted, 5 subnets
O E2 10.10.20.0 [110/20] via 10.10.50.2, 01:55:11, FastEthernet0/5
O 10.10.40.0 [110/2] via 10.10.50.2, 01:55:11, FastEthernet0/5
C 10.10.50.0 is directly connected, FastEthernet0/5
O 10.10.40.0 [110/2] via 10.10.50.2, 01:55:11, FastEthernet0/5
C 10.10.70.0 is directly connected, FastEthernet0/2
172.168.0.0/29 is subnetted, 7 subnets
O 172.168.0.0 [110/3] via 10.10.50.2, 01:55:11, FastEthernet0/5
O E2 172.168.0.8 [110/20] via 10.10.50.2, 01:55:11, FastEthernet0/5
C 172.168.0.16 is directly connected, Vlan0
O 172.168.0.24 [110/2] via 10.10.50.2, 01:55:11, FastEthernet0/5
C 172.168.0.32 is directly connected, Vlan0
C 172.168.0.40 is directly connected, Vlan0
C 172.168.0.48 is directly connected, FastEthernet0/4
192.168.0.0/29 is subnetted, 1 subnets
O 192.168.0.0 [110/3] via 10.10.50.2, 01:55:01, FastEthernet0/5
192.168.1.0/29 is subnetted, 1 subnets
O 192.168.1.0 [110/4] via 10.10.50.2, 01:55:01, FastEthernet0/5

Switch#
Switch#
Switch#
Switch#

```

Physical

Config

CLI

Attributes

IOS Command Line Interface

```

Type-5 AD External Link States
Link ID ADV Router Age Seq# Checksum Tag
10.10.20.0 1.1.1.1 1285 0x00000004 0x00000000 0
172.168.0.8 1.1.1.1 1285 0x00000004 0x00000000 0

Router#show ip route ospf
10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O E2 10.10.20.0 [110/20] via 10.10.40.2, 01:53:24, GigabitEthernet0/2
O 10.10.40.0 [110/2] via 10.10.40.2, 01:53:24, GigabitEthernet0/2
O 10.10.50.0 [110/2] via 10.10.40.2, 01:53:24, GigabitEthernet0/2
O 172.168.0.0/29 is subnetted, 7 subnets
O E2 172.168.0.8 [110/20] via 10.10.40.2, 01:53:24, GigabitEthernet0/2
O 172.168.0.24 [110/2] via 10.10.40.2, 01:53:24, GigabitEthernet0/2
O 172.168.0.32 [110/3] via 10.10.40.2, 01:53:24, GigabitEthernet0/2
O 172.168.0.40 [110/3] via 10.10.40.2, 01:53:24, GigabitEthernet0/2
O 172.168.0.48 [110/3] via 10.10.40.2, 01:53:24, GigabitEthernet0/2
O 192.168.1.0/29 is subnetted, 1 subnets
O 192.168.1.0 [110/2] via 192.168.0.2, 01:53:34, GigabitEthernet0/1

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
NI - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O E2 10.10.20.0/30 [110/20] via 10.10.40.2, 01:54:39, GigabitEthernet0/2
O 10.10.40.0/30 [110/2] via 10.10.40.2, 01:54:39, GigabitEthernet0/2
O 10.10.50.0/30 [110/2] via 10.10.40.2, 01:54:39, GigabitEthernet0/2
C 10.10.40.0/30 is directly connected, GigabitEthernet0/2
L 10.10.40.1/32 is directly connected, GigabitEthernet0/2
L 10.10.70.0/30 is directly connected, GigabitEthernet0/0
L 10.10.70.1/32 is directly connected, GigabitEthernet0/0
172.168.0.0/29 is subnetted, 7 subnets
O 172.168.0.0/29 [110/3] via 10.10.40.2, 01:54:39, GigabitEthernet0/2
O E2 172.168.0.8/29 [110/20] via 10.10.40.2, 01:54:39, GigabitEthernet0/2
S 172.168.0.16/29 [110/2] via 10.10.70.2
O 172.168.0.24/29 [110/2] via 10.10.40.2, 01:54:39, GigabitEthernet0/2
O 172.168.0.32/29 [110/3] via 10.10.40.2, 01:54:39, GigabitEthernet0/2
O 172.168.0.40/29 [110/3] via 10.10.40.2, 01:54:39, GigabitEthernet0/2
O 172.168.0.48/29 [110/3] via 10.10.40.2, 01:54:39, GigabitEthernet0/2
192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
O 192.168.0.0/29 is directly connected, GigabitEthernet0/1
L 192.168.0.1/32 is directly connected, GigabitEthernet0/1
O 192.168.1.0/29 is subnetted, 1 subnets
O 192.168.1.0/29 [110/2] via 192.168.0.2, 01:54:45, GigabitEthernet0/1

Router#
Router#
Router#
Router#

```

Show ip route for the muli layer switch that servers as a bridge

```
IOS Command Line Interface

10.10.40.1 172.168.0.1 1275 0x80000004 0x00c2bc
10.10.50.1 172.168.0.51 1271 0x80000004 0x00a1e2
10.10.60.1 192.168.0.1 1267 0x80000004 0x00e3c7

Link ID Type-5 AD External Link States
ADP Router Age Seq# Checksum Tag
10.10.20.0 1.1.1.1 1313 0x80000004 0x00bbd9 0
172.168.0.8 1.1.1.1 1313 0x80000004 0x007fe5 0

Switch#show ip route ospf

  10.0.0.0/30 is subnetted, 5 subnets
    10.10.70.0 [110/2] via 10.10.60.1, 01:53:15, FastEthernet0/4
    10.10.70.0 [110/2] via 10.10.50.1, 01:53:15, FastEthernet0/5
  172.168.0.0/29 is subnetted, 7 subnets
    O 172.168.0.0 [110/2] via 10.10.40.1, 01:53:15, FastEthernet0/6
    O 172.168.0.16 [110/2] via 10.10.50.1, 01:53:15, FastEthernet0/5
    O 172.168.0.32 [110/2] via 10.10.50.1, 01:53:15, FastEthernet0/5
    O 172.168.0.40 [110/2] via 10.10.50.1, 01:53:15, FastEthernet0/5
    O 172.168.0.48 [110/2] via 10.10.50.1, 01:53:15, FastEthernet0/5
  192.168.0.0/29 is subnetted, 1 subnets
    O 192.168.0.0 [110/2] via 10.10.40.1, 01:53:15, FastEthernet0/4
  192.168.1.0/29 is subnetted, 1 subnets
    O 192.168.1.0 [110/3] via 10.10.60.1, 01:53:15, FastEthernet0/4

Switch#show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       F - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/30 is subnetted, 5 subnets
    C 10.10.20.0 is directly connected, FastEthernet0/7
    C 10.10.40.0 is directly connected, FastEthernet0/6
    C 10.10.50.0 is directly connected, FastEthernet0/5
    C 10.10.60.0 is directly connected, FastEthernet0/4
    O 10.10.70.0 [110/2] via 10.10.60.1, 01:55:17, FastEthernet0/4
      [110/2] via 10.10.50.1, 01:55:17, FastEthernet0/5
  172.168.0.0/29 is subnetted, 7 subnets
    O 172.168.0.0 [110/2] via 10.10.40.1, 01:55:17, FastEthernet0/6
    R 172.168.0.8 [120/1] via 10.10.20.1, 00:00:11, FastEthernet0/7
    O 172.168.0.16 [110/2] via 10.10.50.1, 01:55:17, FastEthernet0/5
    C 172.168.0.24 is directly connected, FastEthernet0/1
    O 172.168.0.32 [110/2] via 10.10.50.1, 01:55:17, FastEthernet0/5
    O 172.168.0.40 [110/2] via 10.10.50.1, 01:55:17, FastEthernet0/5
    O 172.168.0.48 [110/2] via 10.10.50.1, 01:55:17, FastEthernet0/5
  192.168.0.0/29 is subnetted, 1 subnets
    O 192.168.0.0 [110/2] via 10.10.40.1, 01:55:17, FastEthernet0/4
  192.168.1.0/29 is subnetted, 1 subnets
    O 192.168.1.0 [110/3] via 10.10.60.1, 01:55:17, FastEthernet0/4

Switch#
Switch#
Switch#
Switch#
```

4.2)The challenges that BrightSoft Solutions faces with its current mixed RIP and OSPF environment

Routing Protocol Mismatch (Distance-Vector vs. Link-State)

The underlying problem lies in the fact that there's an incompatibility between the two protocols.

Different Philosophies:

RIPv2 (Floor 1): RIPv2 is a distance vector protocol and is better known as “routing by rumor.” This protocol uses only one routing metric to make routing decisions—the hop count. The convergence time is poor as RIPv2 uses periodic full routing updates to its direct neighbors with a hop limit of 15.

OSPF (Floor 2 and Floor 3):

This protocol is link-state. It constructs an accurate map like the one used in GPS of its “area” its whole environment, making routing decisions based on a “cost” usually

calculated from bandwidth, with incredibly rapid convergence via triggered 'minimal updates' called 'Link State Advertisements'.

No Native Communication:

These protocols are like two people speaking different languages. A RIP-speaking router has no idea about the complex topological map that an OSPF-speaking router maintains, and vice versa.

The ASBR (Autonomous System Boundary Router) Requirement:

To get them to communicate with each other, BrightSoft has to set up this router for example, a centralized one that both communicate with to be an ASBR. The ASBR has to be set up to run two different protocols at the same time and has to be configured to perform route redistribution a function that involves translating and then "injecting" routes in one protocol to the other protocol. The route redistribution is invariably causing problems to the network.

The Deep-Dive Challenges of Route Redistribution

It is here that the most serious technical difficulties reside.

Incompatible Metric Translation (Apples to Oranges):

Ensemble RIP-OSPF: If a high-speed OSPF pathway from Floor 2 and 3 is brought into the RIP environment on Floor 1, the static metric values assigned in the OSPF protocol are nullified. A static "seed metric" hop count has to be assigned by the network administrators to bridge the two environments. This is an arbitrary figure for example, "5 hops." A 10Gbps pathway and a 100Mbps pathway would carry the same static metric values if brought into the RIP environment. The Floor 1 RIP routers wouldn't be able to make an intelligent choice.

RIP Injection into OSPF: When the route received from Floor 1's RIP is injected into the OSPF environment in Floor 2, it is advertised as an External Route.

By default, this is an External Type 2 (E2) route. The E2 route has a fixed-cost metric that never varies regardless of your location within the OSPF domain. This causes suboptimal routing since the ASBR router on Floor 2 and 3 may take an unusually long path to get to the ASBR if the E2 metric to every location is the same.'

Administrators can modify this to External Type 1 (E1), which is more intelligent since it adds the internal OSPF cost to the redistributed external routes' cost; however, this process has to be manually configured.

High Risk of Persistent Routing Loops:

Since there needs to be two-way communication in BrightSoft's setup, there has to be “mutual” or “two-way” redistribution RIP injection into OSPF and vice versa. This is notoriously dangerous and “bad practice” unless tightly controlled.

A route learned in OSPF can be injected into RIP, propagate within the RIP domain, and then be learned again by another ASBR and injected “back into OSPF” with resultant “route-poisoning” in RIP or “LSA flooding” in OSPF to blackhole traffic.

These need complex error-prone setups such as route-tagging, route-maps, and prefix-lists to filter the routes that are re-ad

Conflicting Administrative Distance (AD):

A router uses AD to determine whether to “trust” one protocol over another if it hears about the same route from more than one protocol. The default AD value for OSPF is 110. The default AD value for RIP is 120. This means that the router will always prefer the OSPF route over the RIP route. This becomes critical, particularly with the backup route, if there was supposed to be traffic in the RIP network, but then it also finds an inferior route in the OSPF network; it ends up taking the OSPF route simply due to it having lower AD which is more trustworthy.

Administrative and Troubleshooting Complexity

Inconsistent Convergence Times: This contributes to the instability.

OSPF: Convergence time is seconds. When there is a failure in the link, it floods an update; all routers calculate immediately.

Convergence time of RIP: Converges in minutes. It uses “hold-down timers” and “flush timers” with a of maximum 180-240 seconds to time out and delete the dead route.

The Impact: If there is an issue with the link on Floor 1, the OSPF domain in the shared router (floor 2 and 3) will be unaware of the problem for minutes. For this entire period, the “Floor 2 'Shared router will send traffic to the ASBR on the failed route, which in turn black-holes the packets.

Asymmetric Routing and “Broken” Tools:

Redistribution results in asymmetric routing: A packet may come out from the Developer's PC (Floor 1), then traverse the RIP domain and the ASBR to reach the Staging Server (Floor 3). But the return packet might take a “better” OSPF route and follow a wholly different path.

the impact: This affects normal debugging utilities such as traceroute, since traceroute will either fail or give incomplete results. Even more serious, though, it affects stateful firewalls. The firewall interprets the “request” packet that leaves from one interface and then interprets the “response” packet that arrives on another interface. Because the “response” packet has no corresponding “request” packet that sent it, the firewall interprets it as an attack and kills the connection.

Severe Impact on the Failover Link “The backup link between the Shared router and the Business Analysts router and adds to the above problems to make them exponentially worse. It creates a second, Uncontrolled Redistribution Point: The above-mentioned connectivity is referred to as the “back door” that joins the OSPF network to BA's RIP network (BA's router). This results in the creation of the “second” ASBR and “second” Guaranteed Routing Loops: Since there are two different places that routes can be injected from, either from the OSPF-to-RIP boundary or from the RIP-to-OSPF boundary, there is no longer a possibility of creating a closed-loop; rather, it is almost guaranteed. One route can ‘leak’ over one boundary and be injected over the other boundary. Uncontrollable Failover: Stable failover is unachievable with this design. First, which is the prime pathway? Then there is the secondary. As there are two protocols and two boundary routers, the dynamics are non-deterministic. If the primary circuit fails, the network will NOT “failover” smoothly. Instead, both protocols will attempt to reconverge on different timescales with routes flipping aggressively between them. The network might “settle down” on a suboptimal and highly latency-intensive route over the backup circuit and might NEVER come back to the original route even if the underlying problem with the primary circuit was solved. “Failover”-type connections that are supposed to increase resilience end up making the network’s most significant point of failure.

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